# Measuring the Task Value of Precision Upgrades in Closed-Loop Agents

Anonymous ACL submission

## Abstract

Outcome flips do not by themselves show that a low-precision agent needs more bits. A fallback is useful only if its rescues outweigh its damages. We measure this signed task benefit by crossing the source of the current action with the model that continues the interaction. On Qwen3 agents under simulated quantization in $\tau^2$-bench, replacing the first tool-structure-divergent W4 action with BF16 changes 15.5–20.9% of outcomes, yet the pooled net gain is $+0.7$ pp (new-task 95% interval $[-2.1, +3.4]$). Action-only fallback can also harm: a pre-registered telecom holdout has a negative estimate of $-4.6$ pp; its protocol classifies this as inconclusive because it omitted a negative-effect category. A second W4 checkpoint and a semantically identical tool-argument reordering also change outcomes. At shared states, BF16's additional action value over the second checkpoint remains unresolved; equivalence is not established. A frozen uncertainty trigger enriches outcome-changing decisions without a reliable net benefit over random selection. These findings motivate evaluating the signed benefit of the upgrade actually executed, interpreting flips with reference interventions, and selecting decisions by task benefit rather than sensitivity alone.

## 1 Introduction

A changed decision does not by itself justify a higher-precision fallback. When a low-precision LLM agent (Q) can call its BF16 reference (B), the deployment question is whether that call improves the task outcome (Li et al., 2026c). Deciding when to use that fallback requires knowing the task benefit of a specific upgrade: B may **replace the current action, take over from the next step, or do both**. These choices intervene at different points in a closed-loop interaction. Average success rates compare complete agents, while KL and action disagreement describe local drift; neither directly measures what a particular fallback decision gains or loses.

Figure 1a shows a decision from an airline task. Q looks up a reservation id that does not exist; B looks up the user's profile. We execute both actions and continue each with either model. Only BB succeeds. In this case, B's action helps only when B also continues. Evaluating the action alone misses that rescue.

A model that takes over a forked trajectory writes both the next action and the continuation (Gonuguntla, 2026). Their contributions are coupled in that evaluation. We separate them by crossing the action source with the continuing model at an audited, restored state (§3.3). Two reference interventions help interpret the outcome changes: substituting another W4 checkpoint (Q′), and reordering argument keys while preserving the tool operation. We study Qwen3-4B and Qwen3-8B (Qwen Team, 2025) under simulated weight and KV-cache quantization in $\tau^2$-bench (Yao et al., 2024; Barres et al., 2025). Scope checks cover BFCL multi-turn (Patil et al., 2025), ALFWorld (Shridhar et al., 2021), and ordinary sampled states from Mistral-Small-3.2-24B (Mistral AI, 2025) (Appendix F).

**Contributions.** We develop an auditable protocol for measuring three precision interventions on shared states. Building on step values under a continuing policy (Setlur et al., 2024), the protocol combines action–continuation contrasts, paired rescue–damage accounting and two reference substitutions. Our empirical contribution is to separate outcome sensitivity from the task value of precision: substitutions can change outcomes without increasing bit width or changing the executed tool operation. Establishing a need for higher precision therefore requires measuring the additional task benefit of the specified upgrade.

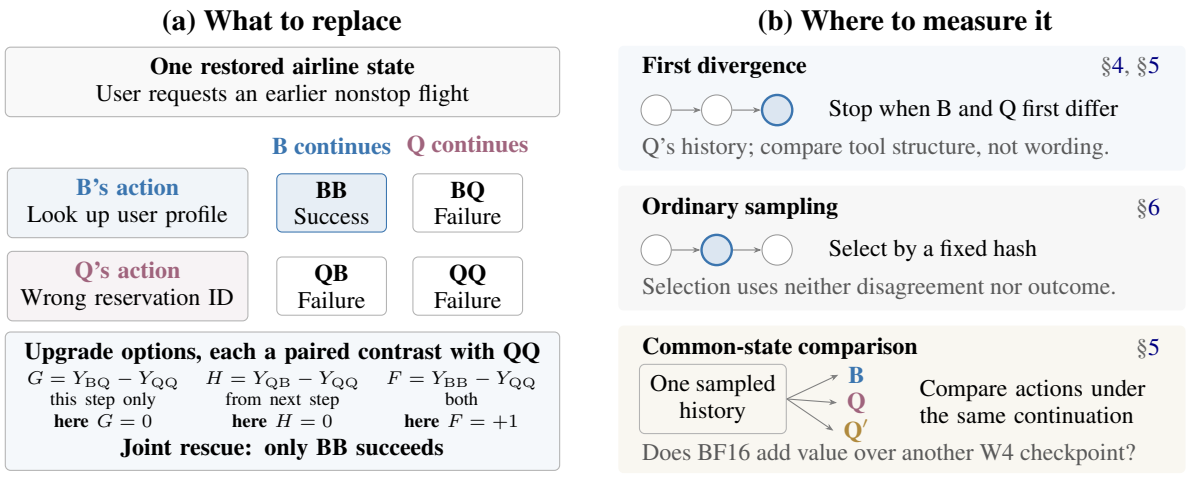



Figure 1: An upgrade has both an intervention and a state population. (a) Cross the current action with the continuing model, then compare each upgrade with QQ. In this airline example (Qwen3-4B, AWQ-style W4; task 31, decision 1 of 12), only joint replacement succeeds. (b) First-divergence states measure consequences at disagreements; ordinary sampling also includes agreements. Comparing multiple action sources on the same sampled history isolates their additional task value. The sampling diagrams are schematic.

Three findings build this argument:

1. **Fallback can rescue and damage tasks.** Replacing the current W4 action produces substantial rescues and damages, which largely cancel in the average. On new telecom tasks, the action-only fallback has a negative estimate. Its frozen protocol classifies the result as inconclusive because it lacks a negative-effect category (§4).

2. **Flips alone do not identify a need for more bits.** A second W4 checkpoint and a semantically identical key-order change also alter outcomes. A direct common-state comparison does not establish either BF16's extra action value or equivalence (§5).

3. **Finding sensitive decisions does not establish net gain.** One frozen Q-only rule finds more outcome-changing decisions than random selection, but no tested rate reliably improves net gain after correction (§6).

## 2 Related Work

**Quantized agents compared episode by episode.** *Flat Score, Amplified Failures* (Jang et al., 2026) is closest in setting: on $\tau^2$-bench retail and telecom, 4-bit agents keep a flat task score while process damage grows, paired tasks flip in both directions, and recovery is equally frequent at both precisions. Its episodes are sampled independently. The failing action and the recovering controller therefore change precision together. We vary them separately at a fixed state. Two-sided flips under compression were first shown on static tasks (Dutta et al., 2024); agent-compression benchmarks report averages (Dong et al., 2025).

**Forks and counterfactual replay.** *The Replay Gap* (Gonuguntla, 2026) forks live SWE-bench Verified trajectories for per-step model switching. Its swap replaces the next action and continuation together, corresponding to our joint intervention. Its control holds the model fixed; ours holds the bit width fixed. Replacing one action at a restored state and letting the original policy continue (our "this step only" contrast) locates decisive errors (Zhang et al., 2025), attributes failures (Shah, 2026) and yields training signal (Li et al., 2026a,b); the factorial design itself is standard.

**Precision routing, escalation and their signals.** DMR (Li et al., 2026c) learns a per-step BF16/W4 router from KL supervision and episode reward; sensitivity has been localised to phases (Lu et al., 2026), task types (Zhang et al., 2026) and high-entropy, high-KL tokens (Lotfi et al., 2026). Agents also defer single decisions when uncertain (Piatrashyn et al., 2026), learn cost-aware escalation (Lei and Huang, 2026), or are routed per step against execution-verified labels (Yang et al., 2026). Continuing another model's trajectory has its own cost (Ganz et al., 2026). Confidence does not reliably locate consequential steps under executed replay (Zhang, 2026). KL measures dis-

agreement magnitude without identifying its direction (Nikolić et al., 2026). Our evaluation measures action-only, continuation-only and joint substitutions on shared states, using rescue–damage accounting and reference interventions to interpret their task value.

**Evaluation noise and user simulators.** Prefix caching changed multi-turn tool-use trajectories in 36.2% of episodes at 16-bit and 75.0% at 4-bit (Patodiya, 2026); agentic evaluations vary from run to run (Bjarnason et al., 2026; Chen et al., 2026) and with the user model (Seshadri et al., 2026; Jain and Sandhu, 2026).

## 3 Measuring the Task Value of an Upgrade

The measurement target is the success gained or lost by executing a specified upgrade from the same state. We separate the action from the continuation so that this target matches the fallback that will actually run. Reference interventions then test what an observed change can be attributed to.

### 3.1 Setting

**Tasks and agents.** We use $\tau^2$-bench in text mode. The main development panel has 58 retail and 27 airline tasks. We also evaluate 80 telecom development tasks and a holdout of 34 telecom tasks. The holdout protocol was fixed before any agent ran those tasks. The agents are Qwen3-4B-Instruct and Qwen3-8B with thinking disabled; a second family, Mistral-Small-3.2-24B, is a scope check (Appendix F). One greedy Qwen3-30B-A3B-Instruct user simulator serves all agents. Retail and airline success requires a final-database match and normal termination. It excludes some official communication checks. Telecom uses the official task-basis success. BFCL multi-turn and ALFWorld have no LLM user; each item is evaluated once (Appendix A).

**Quantization and serving.** Q is a simulated (fake-quantized) checkpoint served with BF16 compute on the same path as B. The main W4 settings use GPTQ (Frantar et al., 2023) or AWQ-style scaling (Lin et al., 2024), with group size 128 and a BF16 KV cache. Calibration uses BF16 agent trajectories from 57 retail and airline tasks, disjoint from the development panel. Separate KV8 configurations quantize the cache to INT8 while retaining BF16 weights. Additional quantization settings test the range of upgrade values (Appendix Table 4). B and Q run on vLLM (Kwon et al., 2023) with batch-invariant kernels (He and Thinking Machines Lab, 2025) and deterministic compilation in separate server processes (no shared prefix cache). A configuration is a model with a quantization setting. Each replicate, called a *block*, runs Q over the task pool with a fresh session-identifier line in the system prompt. A *unit* is one task in one block.

### 3.2 Four branches on one state

**Frozen state.** Before Q's decision $t$, we restore state $s_t$. It contains the full message history, the user simulator's visible conversation, and the used step and error budgets. Replaying the recorded tool calls rebuilds the environment and user-device databases. We save the state before running any branch. B and Q then act from identical prompt token ids.

**Four branches.** Let $Y_{\mathrm{XY}} \in \{0, 1\}$ denote task success when X supplies the current action and Y continues, with X, Y $\in \{\mathrm{B}, \mathrm{Q}\}$. All branches start with the same remaining budget. Each upgrade is compared with QQ: replace this step ($G$), take over from the next step ($H$), or do both ($F$). Figure 1 defines these contrasts. A *rescue* occurs when the upgrade succeeds and QQ fails; a *damage* is the reverse. Their sum is *turnover*, and their difference is *net gain*. These quantities measure task consequences. Additional notation appears in Appendix A.1.

**The action being replaced.** The intervention replaces a complete assistant turn, including its language and structured tool calls. The chosen message and tool results enter the subsequent history; the continuing model makes every later decision. The continuation is part of the intervention: it determines how the substituted action enters the rest of the task.

**Reference interventions.** A large turnover alone cannot distinguish a need for precision from sensitivity to model substitution. We rerun the design with Q′ in place of B. Q′ is another GPTQ-W4 checkpoint of the same model, built by resampling states and activation rows from the same calibration trajectories. The Q episodes, tasks and prompt blocks stay fixed. This reference covers q4 and q8 GPTQ-W4, plus GPTQ-W3 as a stress check. Its protocol was fixed before outcomes were read. Section 5 adds a direct

comparison on shared states and a key-order change of Q's own action.

### 3.3 Making the counterfactuals auditable

The branches differ in the intended intervention, with history, environment and remaining budgets restored before execution. Outcome-independent no-op audits check that restoring a state preserves its original result: all 924 sampled development states and all 1202 holdout divergent states matched. The holdout protocol was fixed before its tasks ran. These checks make the paired comparisons trustworthy under the tested greedy, dense-BF16 serving stack; Appendix A gives the acceptance rules and Appendix C the holdout protocol.

### 3.4 Sampling and inference

**State sources.** A first divergent state is the earliest Q-history decision where B and Q differ in action kind, tool calls or normalised arguments. Wording alone does not qualify. Ordinary states are selected by a fixed hash before action generation, one per task and block. A *common state* is an ordinary sampled state at which several action sources are compared; Figure 1b distinguishes the state sources. Full-pool rates include every task–block unit, counting no divergence as zero effect; mode and pattern analyses condition on divergent states. We write $G_{\text{all}}$ for the full-pool action-only net gain, and $G$ for its average on divergent states. Contrasts are paired within a configuration and descriptive across configurations, which select their own divergences; only complete blocks enter.

**Intervals and interpretation.** The main figures use new-task intervals, which resample both tasks and prompt blocks, and the holdout uses its registered confidence level. Fixed-pool intervals and the adjusted common-state reading appear in the appendix. An interval containing zero leaves the effect unresolved; equivalence requires a separate test. Captions specify the confidence levels. Each branch measures one realised outcome under the specified greedy stack.

## 4 Fallback Can Rescue and Damage Tasks

A fallback can change many outcomes without improving average success, and action-only replacement can lower success on new tasks. Figure 2 separates rescues, damages and their net benefit.

**Large outcome changes, small net estimate.** The development panel has 85 tasks and 20 complete blocks per configuration. We replace Q's action at its first divergence with B's and let Q continue. This changes 15.5–20.9% of outcomes, with rescues and damages of similar size (Figure 2a). No new-task interval of the net excludes zero. Pooled over the four W4 configurations, the full-pool net gain $G_{\text{all}}$ is $+0.7$ pp (new-task 95% $[-2.1, +3.4]$). The net is a local replacement effect, not an end-to-end comparison (Appendix A.2).

**Action-only fallback can harm new tasks.** The holdout tests 34 unseen telecom tasks over 12 fresh blocks and three W4 configurations. Its protocol was fixed before any task ran. Replacement again produces both rescues and damages (Figure 2b). The primary full-pool net gain $G_{\text{all}}$, $-4.57$ pp (crossed 90% $[-7.17, -1.96]$), lies below zero: action-only replacement reduced success on this holdout. Its cause is unknown. The frozen readings had only positive-gain and small-effect categories, a design omission, so the protocol classifies the result as *inconclusive* (Figure 2b; Appendix C). Both pre-registered secondary checks passed: rescues and damages were each positive, and task fragility repeated across independent prompt blocks (Appendix C). Full baseline replay rules out a restoration mismatch at the tested states. The holdout changes domain as well as tasks (Appendix C.1).

**The intervention defines which rescues count.** Figure 2c reports all three modes on the W4 and KV8 divergent-state pools. For W4, $G$, $H$ and $F$ average $+0.5$, $+0.2$ and $+0.7$ pp, respectively; all three new-task intervals cover zero. Similar averages still hide different successes. Joint rescue, where only changing both succeeds, occurs on 4.0% of W4 divergent states. Rescue undone, where B's action succeeds with Q continuing but fails with B continuing, occurs on 4.8%. These are observed shares, not evidence of enrichment over a permutation reference (Appendix B.1). Action-only evaluation misses the former; joint-replacement evaluation misses the latter. Thus a measured gain for one mode cannot stand in for another.

The RTN-W3 scope check illustrates this distinction in aggregate: action-only replacement has a near-zero estimate, whereas continuation takeover and joint replacement have positive gains

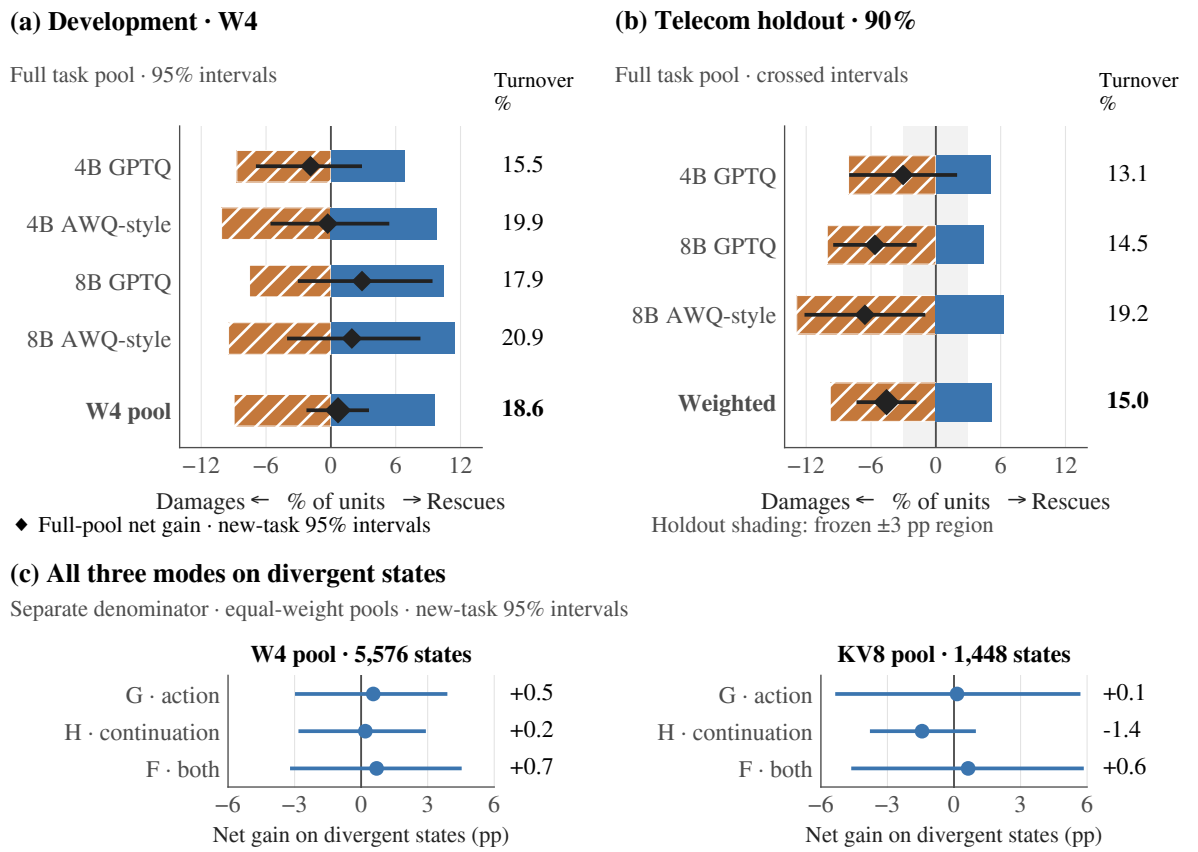



Figure 2: Three upgrade modes and their task consequences. (a,b) Action-only replacement: blue rescue and hatched orange damage rates; diamonds show full-pool net gain $G_{\text{all}}$. All task–block units enter these rates, including units without a divergence. (a) Development: new-task 95% intervals. (b) Holdout: crossed 90% intervals; shading marks the frozen ±3 pp region. Bold rows are pooled or primary. (c) On divergent states only, $G$, $H$ and $F$ compare action-only, continuation-only and joint replacement with QQ. These exploratory pooled estimates use new-task 95% intervals and a different denominator from $G_{\text{all}}$. AWQ means AWQ-style scaling. The holdout is pre-registered; mode and pattern analyses are exploratory. Other intervals and pattern references are in Appendix B.

(Appendix F). This configuration was selected from five exploratory stress settings; it shows why evaluating only the current action cannot settle the value of a takeover.

These mode-specific values answer a question that a single success-rate comparison cannot: what happens when this action is changed and this controller handles the consequences? A beneficial complete-agent substitution does not imply that borrowing one of its actions will help. Conversely, an action-only failure does not settle whether taking over the continuation would be useful.

## 5 Flips Alone Do Not Identify Precision Needs

An outcome can flip without either more bits or a different tool operation. A flip therefore shows sensitivity to substitution. To ask whether it indicates a need for higher precision, we compare BF16 with two reference interventions (Figure 3). Each reference answers a different question on its own state population.

**A same-precision checkpoint also changes outcomes.** Q′ replaces the current action without increasing bit width. On the matched W4 task–block panel it produces 17.7% turnover, compared with 16.8% for BF16, with rescues and damages in both directions (Figure 3a). The W3 stress comparison also shows bidirectional changes (Appendix D.1). These are observed turnover estimates; equivalence within ±3 pp is not established.

**An identical tool operation can lead to a different outcome.** At 300 eligible Qwen3-8B states, we reverse Q's tool-argument key order. The

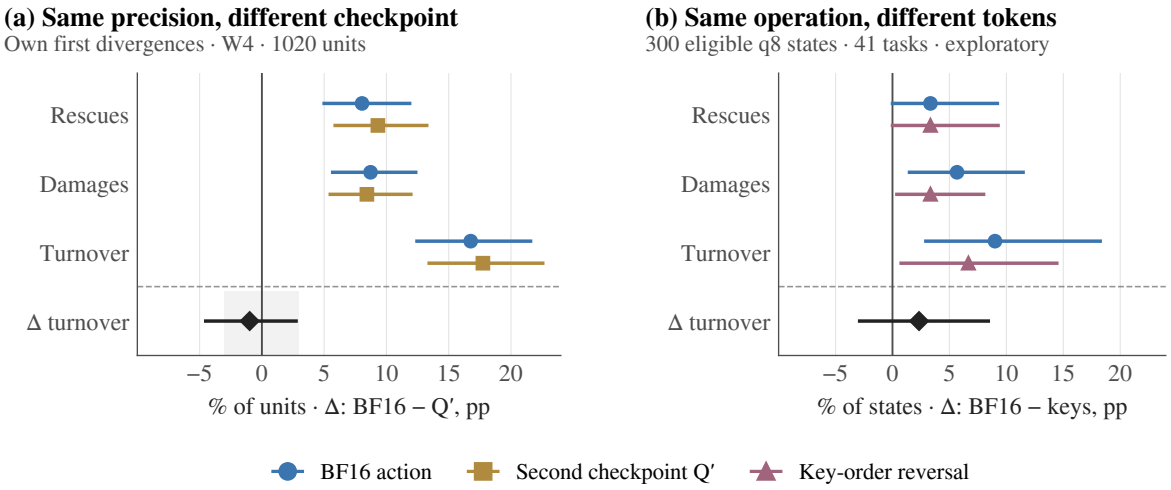



Figure 3: Two reference interventions change outcomes without increasing bit width. (a) A second W4 checkpoint and BF16 are substituted at their own first divergences; lines are new-task 95% intervals. (b) Reversing argument keys preserves the tool operation; lines are the wider of block and new-task 95% intervals on the eligible states. Paired turnover differences are consistently BF16 minus the reference and are separated from rates by a dashed line. The populations differ and the reference rates are not subtracted as a noise floor. The direct shared-state value comparison and all other intervals appear in Appendix D.2 and Table 12.

call still parses to the same arguments and produces the same environment effects. Only the token sequence read by the agent changes. Outcomes change on 6.7% of these states, versus 9.0% when replacing the action with BF16's (Figure 3b). The paired turnover difference (BF16 minus key reversal) is $+2.3$ pp $[-2.9, +8.4]$, leaving the difference unresolved. This exploratory reference covers one representation change on 41 tasks, rather than a general perturbation distribution (Appendix D.3).

**Shared states isolate the additional value of BF16.** The same-precision panel selects each substitute's own first divergence. It compares two selection-and-replacement protocols, rather than two actions at the same decision. We therefore compare B and Q′ directly at 510 shared ordinary states from Qwen3-8B BF16 histories. The extra action value is $+1.0$ pp with Q continuing and $+0.8$ pp with B continuing. Both remain unresolved, and equivalence is not established. Appendix D.2 gives the adjusted intervals and the reading rule adopted after an unresolved first look. The experiment neither establishes BF16 as better nor rules out a useful advantage.

Together, the references establish a limit on interpreting flips: a changed outcome does not identify a decision that needs higher precision. The additional task value of BF16 requires a direct paired comparison of actions under the continuation that will run.

## 6 Sensitivity Enrichment Does Not Establish Net Gain

The frozen uncertainty rule finds decisions whose outcomes change, but it does not reliably select a net benefit. Figure 4 tests these two objectives separately.

**Selecting outcome-changing decisions.** $G$, $H$ and $F$ require running B. We test selection under a Q-only constraint: the rule uses no BF16 computation before choosing an upgrade. We choose and freeze the rule on development data, then evaluate it on later prompt blocks. On one development block, we choose among entropy, margin and mean log-probability by net gain over random selection. We freeze the winning margin rule before validation: at low minimum token margin, B supplies the action and Q continues. This implements $G$; margins are known to predict cross-precision divergence (Du et al., 2026). Per-configuration thresholds target 5, 10 and 20% of development states. Validation can select a different fraction, so random and oracle baselines match the actual counts (Figure 4).

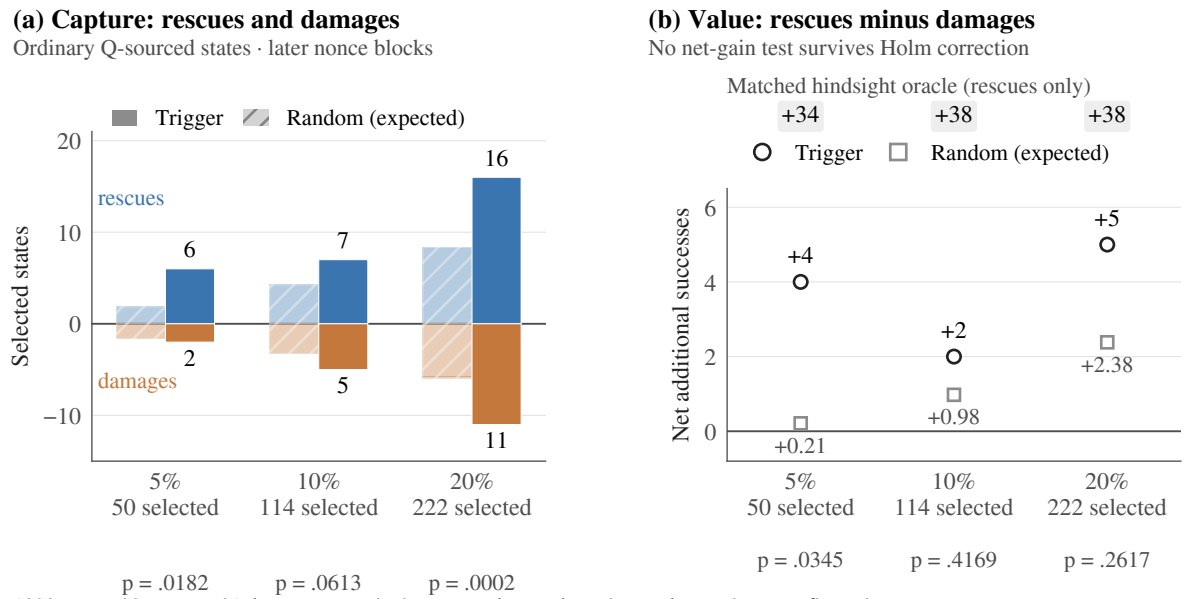



Figure 4: One frozen Q-only margin trigger on 1020 ordinary Q-sourced validation states. (a) Selected rescues and damages versus matched-count random selection. (b) Net additional successes; the grey strip gives the matched-count hindsight oracle, which selects rescues by construction. Nominal rates are shown with actual selected counts. Exact one-sided $p$-values compare capture in (a) and net gain in (b); no net test survives Holm correction across the three rates. Full values: Table 13.

We test on two later blocks of the same development tasks, separate from telecom holdout. The states come from Q's own histories. Six configurations (four W4 and two KV8) yield 1020 states, including 69 outcome changes. At the 20% nominal rate, the rule captures 27 changed outcomes against 14.30 expected from matched random selection ($p = .0002$). The other rates and a secondary test on BF16 histories appear in Appendix E.1.

**Testing benefit against random selection.** The captured states were rescues and damages alike (Figure 4a). None of the three rates shows a reliable net benefit over matched random selection after Holm correction (Figure 4b). No rate was designated primary; exact tests and the nominal 5% result appear in Appendix E.2. A hindsight oracle with the same per-configuration counts nets far more (Figure 4b), positive by construction because it picks only rescues.

**What this test establishes.** Enriching outcome-changing decisions is not enough for reliable net gain (cf. Zhang, 2026). The later blocks reuse the development tasks, and only this simple frozen rule is tested. Learned and other uncertainty-based escalation rules require their own net-benefit tests (Li et al., 2026c; Piatrashyn et al., 2026; Lei and Huang, 2026).

| Setting | States | Flip % | Task-value reading |
|---|---|---|---|
| BFCL | Divergent | 5.2–15.2 | Pooled net +0.1; unresolved |
| ALFWorld | Divergent | 21.6–24.6 | Pooled net −3.7; unresolved |
| Mistral 24B | Sampled B | 10.0–10.4 | All modes unresolved |
| RTN-W3 | Divergent | 6.7 | Joint $F = +18.4$ [9.4, 28.2] pp |

Table 1: Scope checks on separate populations. Divergent states are first tool-structure disagreements; sampled states use BF16 histories. Flip rates are action-only. RTN-W3 is an exploratory 8B example selected from five stress configurations; its interval is crossed 95%. Full populations and intervals: Appendix F.

## 7 Implications for Precision Fallback

A precision upgrade should be judged by how many tasks it saves minus how many it spoils. This signed objective is distinct from agreement with BF16 and from the number of changed outcomes. The reference experiments make that distinction testable.

**Match the label to the executed mode.** The three upgrade modes spend BF16 computation at different points. Replacing one action lets Q handle its consequences. Taking over the continuation preserves Q's current action. Joint replacement changes both. A selector's outcome target should match the mode it will execute. A label for joint takeover can count a success that action replacement cannot obtain. Conversely, a success under Q's continuation can disappear under B's. These patterns make the evaluation target consequential even when average gains are similar. BF16 also need not be the better end-to-end policy: Qwen3-4B GPTQ-W4 achieves 40.0% end-to-end DB success versus BF16's 35.8% (Appendix A.2).

**Optimise benefit, not response to intervention.** Our frozen margin rule finds outcome changes on later blocks of the same tasks. It captures rescues and damages together, giving no reliable net benefit over matched random selection. The finding concerns this frozen rule. The hindsight oracle identifies available gains after outcomes are known; it does not establish that Q-visible features can predict them. Learned routing should be evaluated on held-out decisions by signed task benefit. Cost-aware comparisons should also match BF16 computation, because continuation takeover can use more decisions than a single-action replacement.

**From realised consequences to deployment value.** Exact replay makes a comparison auditable under fixed execution conditions. It does not make that setting representative of deployment. Key-order reversal demonstrates that individual labels can depend on the representation the agent reads. To estimate expected benefit, specify a distribution of prompts and user conditions, repeat paired branches across it, and average the success differences. Average benefit and individual label stability are separate questions. Our task-level intervals address task sampling; they do not cover every change to the user model or serving stack. On the telecom holdout, the action-only estimate indicates a net loss in the tested setting, although the frozen protocol has no category for it. Because domain and tasks change together, these data do not identify the cause or establish a general domain-transfer effect.

**What the references change.** A turnover statistic combines beneficial and harmful interventions. It cannot say which side dominates, or whether the change required more bits. The same-precision checkpoint answers the latter question by supplying a different action without changing bit width. Key reordering goes further: the executed tool operation stays fixed, yet the later outcome can change. These observations limit an attribution from a flip to a precision deficit. They do not establish that BF16 has no additional value. That stronger question requires the shared-state value comparison, whose uncertainty remains substantial here.

**A concrete evaluation target.** A router trained to predict disagreement can succeed on that target while failing to save tasks. Our trigger experiment evaluates both quantities on the same selected states, so enrichment cannot substitute for the benefit test. For an action-only router, the desired target is the task success difference after Q handles the substituted action. For a takeover router, it is the difference after B handles the continuation. A task-saving policy must avoid damaging successes that Q would have retained, so evaluation must count both directions.

**Coverage and practical use.** Outcome changes and signed benefit are also measured without an LLM user and in a second model family (Table 1). BFCL and ALFWorld remove the user simulator; Mistral changes the model family. The RTN-W3 example shows that a degraded agent can benefit from takeover even when action-only replacement is unhelpful. The shared lesson is to measure the executed upgrade's signed benefit on its intended state population. Appendix F reports the paired estimates and controls.

## 8 Conclusion

Precision fallback should be evaluated by the task benefit of the upgrade actually executed. Our audited forks distinguish action replacement, continuation takeover and joint replacement. The W4 panel shows substantial rescues and damages with little net gain. Same-precision and representation references establish why flips alone cannot diagnose a need for more bits. A useful selector must identify signed benefit under the intended continuation.

## Limitations

**Inference scope.** The contrasts measure realised task effects under the fixed execution stack. Fixed-pool and new-task intervals address different targets; intervals containing zero leave an effect unresolved, while equivalence requires the designated equivalence test. $G$, $H$ and $F$ are offline intervention values rather than the gain of a deployed selection policy. The rescue-pattern and label-permutation analyses are exploratory and post hoc (Appendix B.1).

**Two model families, limited coverage.** Most crossed forks use Qwen3 models: 4B and 8B throughout, plus a single Qwen3-14B configuration on BFCL. The second family, Mistral-Small-3.2-24B (GPTQ-W4, RTN-W4), was forked only at ordinary BF16-sourced $\tau^2$-bench states of an 83-task conditional population (Appendix F.3); it has no first-divergent-state forks, no same-precision reference and no outcome-class analysis. Effects may differ at other scales and in other families. End-to-end runs also exist for Qwen3-14B and Qwen3-32B; apart from that one Qwen3-14B configuration they have no crossed forks, and we draw no conclusion from them (Appendix A.3). A time-boxed Llama-3.1-8B-Instruct check (GPTQ-W4, BFCL multi-turn) was stopped at its gate: BF16 solved 5 of 120 gate items, so the four arms carried almost no outcome variance, and it yields no evidence for that family.

**Simulated quantization.** Quantized checkpoints are dequantized and served with BF16 compute on a deterministic path. Deployed low-bit kernels change the numerics and add serving variation that we deliberately remove. We make no claims about latency, memory or deployed kernels, and the cost of a BF16 fallback is outside our scope.

**One user simulator; two small controls.** All $\tau^2$-bench episodes use a single greedy simulator, and continuation effects include its replies. Both second-simulator controls replace only the user after a fixed history, cover 78 and 80 states, and say nothing about real users. The first-divergent-state control has 20 states per configuration, too few to confirm or refute few-pp contrasts. Neither control can be reproduced draw for draw: the replacement user is a hosted Sonnet 5.5 model sampled with its default settings, without temperature or seed control. In the ordinary-state control, Claude Code subagents answered many relay requests in one context, so a responder could see other arms or replicates of the same conversation; the first-divergent-state control removed this exposure by issuing one isolated, single-request call per user turn. The same-precision reference (§5) covers q4 and q8 GPTQ-W4 on the development panel (six blocks) and GPTQ-W3 on three blocks, and swaps one calibration sample for another, re-drawn from the same calibration trajectories; it does not cover checkpoints calibrated on independent datasets. Six is the longest complete common prefix under the frozen rule, because the seventh block of the same-precision q8 design is incomplete. It is our only reference for how much turnover and continuation dependence a change of model without a change of bit width produces, and it says nothing about AWQ, KV8, RTN or rotated KV4.

**Small confirmation and partial state sets.** The pre-registered holdout covers 34 telecom tasks and three configurations. Its pre-registered readings had only positive-gain and small-effect categories; omitting a negative-effect category was a design omission, and it leaves the protocol able to classify the negative estimate only as inconclusive. It applies only to telecom, these three configurations and this simulator. A post hoc replay of all 1202 divergent states reproduced every original QQ episode (Appendix C); it rules out a replay artifact, not other causes of the negative estimate. Most development results come from retail and airline, so the holdout confounds a domain shift (telecom) with generalisation to new tasks. On 80 telecom development tasks, the two q8 holdout configurations show no detectable net effect (Appendix C.1), but their 6-block intervals are wide and q4 has no telecom development forks. These data cannot separate a telecom domain effect from heterogeneity among the 34 holdout tasks. First-divergent-state forks intervene only at the first tool-structure divergence per task and block. Ordinary-state forks sample one state per task and block; their main analyses use three blocks, so their block intervals rest on two degrees of freedom, and the later panels use two to six blocks. Prompt nonces define replicate blocks; they are a researcher-chosen perturbation, not a deployment distribution.

## Ethics Statement

This work involves no crowdworkers, no human annotation and no personal data. Where a user is present ($\tau^2$-bench), it is simulated by an LLM; tasks are public benchmark tasks, and all agents are openly released Qwen3, Mistral-Small-3.2 and (gate check only) Llama-3.1 checkpoints, or simulated-quantized versions of them. Outcomes use the automatic task-success endpoints defined in §3.1, including the database-only retail and airline checks, and action categories are assigned by fixed rules; reported values map to committed analysis outputs (Appendix A). The main risk we see is over-reading. A small forked-state gain on one simulator and a fixed task pool does not certify a quantized agent for deployment, and our end-to-end comparisons do not establish equivalence. Deployment decisions in consequential settings should not rest on these measurements alone.

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# A Measurement Setup and Replay Checks

This appendix supports §3: it gives the splits, state restoration, replay checks and notation needed to reproduce the contrasts.

**Evidence provenance.** Reported values map to committed analysis outputs and frozen manifests. The supplement retains the complete primary comparisons, acceptance checks and protocol amendments needed to assess the paper's claims.

**Guide to the supplementary evidence.** Appendix A specifies the measurement and its replay checks. Appendix B expands the three upgrade modes, the illustrative rescue patterns and repeatable task fragility. Appendix C documents the independent holdout and full baseline replay. Appendix D supplies the precision-reference comparisons and adaptive-analysis timeline. Appendix E supports the frozen selection rule. Appendix F tests the scope stated in Table 1. Each section gives the checks needed to interpret its corresponding main-text claim.

**Splits.** The development split has 58 retail, 27 airline and 80 telecom tasks. The holdout has 34 telecom tasks, and its pre-registration hash is recorded in the split file. The telecom split was fixed from $\tau^2$-bench's 114-task base split by a stratified, salted hash before any run. The external benchmarks use BFCL multi-turn (base, miss_func, miss_param and long_context, 197 items each) and ALFWorld (134 out-of-distribution games). Items were selected by hash in manifests committed before any model run.

**Success endpoints.** For retail and airline we use the final-database match with normal termination. This is not $\tau^2$-bench's official reward, which additionally scores natural-language or communicate checks. Telecom uses the official deterministic task-basis success. Context overflow and unparseable output count as failures, except for the explicitly amended missing-data rules of the Mistral panel (Appendix F.3). Infrastructure outages are missing, never failures, and a block with a missing unit leaves the primary analysis.

**Fork mechanics.** The forced action is returned by a fork agent for the one step that follows $h_t$. A tool call is executed by the real environment, and text is answered by the real simulator, which never reuses observations from the original trajectory. $\tau^2$-bench resets step and error counters when it starts from a history. The fork restores them to the values used at $t$, so the remaining budget equals the original. Every max_steps and too_many_errors episode on the checked blocks is reproduced by the recount.

**Replay acceptance details.** In the engineering gate, one B→B field, the official reward, was missing from older trajectories and was recomputed from the stored reward information before the verdict. Besides the 924 sampled development states, 328/328 sampled Bank U states passed the QQ no-op, and the Bank U gate passed a further 135/135 no-ops (a different set from the holdout's 135 QQ no-ops, all of which also passed; a later full replay of all 1202 holdout divergent states also matched; Appendix C). Serving in separate processes closes the prefix-cache channel (Patodiya, 2026). The same-precision run's acceptance counts are in Appendix D.

**Determinism of the serving stack.** Closed-loop determinism pairs, the same configuration on two nodes at different times, reproduced whole episodes token for token for dense BF16 paths at tensor parallelism 1 and 2 with 1 to 16 concurrent requests (e.g. 257/257 and 1520/1520 episodes). Nine arms that two studies ran independently at the same states were byte-identical. We did not test exactness under KV-cache preemption, or on other models, GPUs or vLLM versions. A deployed AWQ kernel path with fp16 dequantization was not replay-exact, and no reported outcome uses it.

**User simulator.** The served user simulator is Qwen3-30B-A3B-Instruct in every episode; a user-model label stored in the fork state files is outdated and should be ignored.

## A.1 Internal names and notation

**Glossary.** The main text uses plain terms; this supplement also uses the internal names of the analysis code. A *first divergent state* is a *Bank P* state. An *ordinary sampled state* is a *Bank U* state; its sub-banks are drawn from Q's own trajectories (U1) or from trajectories shared by all configurations of a model (U2B: BF16 trajectories; U2Q: Q trajectories). A *common state* is an ordinary state of the common-state design (pre-registered at three blocks, extended to six by amendment),

at which B, Q and Q′ each supply the current action. A *cell* is a configuration, named by model (q4: Qwen3-4B; q8: Qwen3-8B; q14: Qwen3-14B; m24: Mistral-Small-3.2-24B) and quantization setting. Nonce blocks carry ids such as fNN (development) and qN (stress and W3 panels); the holdout blocks have their own ids. Branches are written by action source and continuing model (BB, BQ, QB, QQ; KQ is the key-order branch, Appendix D.3). Symbols: $r$ and $d$ are the rescue and damage rates of the action swap, $M = r + d$ is the turnover and $G_{\text{all}} = r - d$ the net gain on the full pool; $\Delta_{\text{Q}}$, $\Delta_{\text{B}}$ and the off-diagonal share are defined below; $I = \Delta_{\text{Q}} - \Delta_{\text{B}}$ is the interaction of action and continuation (Table 6); $D_{cc} = Y_{\text{QB}} - Y_{\text{BB}}$ compares methods on shared ordinary states (Appendix F.5); $K(c)$ is B's action value beyond Q′'s at a common state under continuation $c$ (Appendix D.2); $n_{\text{blk}}$ is the number of complete blocks.

**Action swap, rescues and damages.** The paired effect of the action swap under each continuation is $\Delta_{\text{Q}} = Y_{\text{BQ}} - Y_{\text{QQ}}$ $(= G)$ and $\Delta_{\text{B}} = Y_{\text{BB}} - Y_{\text{QB}}$, each in $\{-1, 0, +1\}$. When $\Delta_{\text{Q}} \neq \Delta_{\text{B}}$ ("off-diagonal"), the swap's value depends on the continuation. For the action swap, a *rescue* is $(Y_{\text{BQ}}, Y_{\text{QQ}}) = (1, 0)$ and a *damage* is $(0, 1)$; with rates $r$ and $d$, the turnover is $M = r + d$ and the net gain $r - d$. All contrasts measure effects on task success, not which action is correct.

**Same-precision notation.** $M'$, $r'$, $d'$ and the off-diagonal share are defined likewise with Q′ for B.

**Which states.** *Bank P*: for each task, block and cell, the first decision at which $a_{\text{B}}$ and $a_{\text{Q}}$ differ in kind (tool call, text, stop), tool name, call count or order, or normalised arguments; differences in text only are skipped. *Bank U*: one ordinary decision per task, drawn by a hash fixed before generation, in three nonce blocks (up to six for U2B in §5–§7), from Q's own trajectories (U1) or from trajectories shared by all cells of a model (U2B: BF16; U2Q: Q).

**Denominators and inference.** We keep three denominators separate. (1) *Full pool*: every task×block unit, units without a Bank P state counting as zero; $r$, $d$, $M$ and $G_{\text{all}} = r - d$ use it. (2) *Divergent states*: the Bank P states of a cell; $G$, $H$, $F$ and the off-diagonal share use it. (3) *Bank U*: one state per task. Within a cell all branches share each state and all tasks of a block share its nonce, so within-cell contrasts are paired and identified; cross-cell comparisons are descriptive, because each cell selects its own divergences. With $n_{\text{blk}}$ complete blocks (only complete blocks enter), the primary *fixed-pool* interval treats the task pool as fixed (block bootstrap if $n_{\text{blk}} \geq 10$, else block-$t$, a $t$ interval over block means); the *new-task* interval is a crossed task×block bootstrap ("crossed"). We call a contrast *resolved* under an interval when that interval excludes zero and name the interval; "not resolved" means the reported interval or intervals include zero. Where a frozen protocol sets its own reading rule, we state it with the result.

### A.2 Separating local value from end-to-end success

The replacement gains $G_{\text{all}}$, $G$, $H$ and $F$ are not end-to-end comparisons. Table 2 gives the end-to-end comparison, computed separately on the same checkpoints, the same tasks and the same 20 nonce blocks as the forks.

### A.3 Coverage Matrix

Table 3 lists the completed crossed-intervention coverage behind every Qwen3 Bank P and Bank U result; the coverage of the U2B panels (§6, §7) is in Appendix F.5. It counts tasks, states and complete blocks after the strict admission rule. Stress settings are reported separately from the main W4 population.

### A.4 Quantization configurations

Table 4 gives the quantization settings of every checkpoint configuration, read from the checkpoint metadata and the quantization code. Not recorded: the AWQ-style row-sample size (4096 is the code default) and the KV-cache scale dtype (the hook computes in fp32); Mistral has no second calibration draw.

| Cell | Q / BF16 (%) | Q−BF16 [95% CI] | NI 3 pp | TOST ±3 pp |
|---|---|---|---|---|
| q4 GPTQ-W4 | 40.0 / 35.8 | $+4.2$ [$-2.4$, $+10.5$] | pass | fail |
| q4 AWQ-style W4 | 36.8 / 35.8 | $+0.9$ [$-5.8$, $+7.5$] | fail | fail |
| q8 GPTQ-W4 | 30.1 / 30.4 | $-0.2$ [$-7.7$, $+6.8$] | fail | fail |
| q8 AWQ-style W4 | 31.5 / 30.4 | $+1.2$ [$-5.3$, $+7.6$] | fail | fail |
| pool, all four | | $+1.5$ [$-2.7$, $+5.6$] | pass | fail |

Table 2: End-to-end closed-loop success, quantized minus BF16 (pp). Units are paired on nonce, and the crossed task×block bootstrap is drawn jointly across cells. Non-inferiority (NI) at a 3 pp margin passes for q4 GPTQ-W4 and the all-four pool. Equivalence (TOST at ±3 pp) passes nowhere. These comparisons therefore support neither a difference nor equivalence.

| Cell | $\tau^2$-bench retail (Bank P) | $\tau^2$-bench airline (Bank P) | telecom dev (Bank P) | Bank U (retail / airline) | BFCL / ALFWorld |
|---|---|---|---|---|---|
| q4 GPTQ-W4 | 58 t / 727 st<br>20/20 blocks | 27 t / 375 st<br>20/20 blocks | holdout only | 3 blocks<br>393 / 177 rows | 788 items · 573 st / – |
| q4 AWQ-style W4 | 58 t / 986 st<br>20/20 blocks | 27 t / 453 st<br>20/20 blocks | – | 3 blocks<br>393 / 177 rows | 788 items · 642 st / – |
| q4 KV8 | 58 t / 412 st<br>22/22 blocks | 27 t / 181 st<br>22/22 blocks | – | 3 blocks<br>393 / 177 rows | – / – |
| q8 GPTQ-W4 | 58 t / 1017 st<br>20/20 blocks | 27 t / 456 st<br>20/20 blocks | 80 t / 477 st<br>6/6 blocks ; holdout | 3 blocks<br>393 / 177 rows | 788 · 533 st / 134 · 115 st |
| q8 AWQ-style W4 | 58 t / 1087 st<br>20/20 blocks | 27 t / 475 st<br>20/20 blocks | 80 t / 480 st<br>6/6 blocks ; holdout | 3 blocks<br>393 / 177 rows | 788 · 548 st / 134 · 119 st |
| q8 KV8 | 58 t / 585 st<br>22/22 blocks | 27 t / 270 st<br>22/22 blocks | – | 3 blocks<br>393 / 177 rows | – / – |
| q4 GPTQ-W3 | 58 t / 162 st<br>3/3 blocks | 27 t / 70 st<br>3/3 blocks | – | – | 591 · 540 st† / – |
| q4 KV4-rot | 58 t / 164 st<br>3/3 blocks | 27 t / 69 st<br>3/3 blocks | – | – | – / – |
| q8 GPTQ-W3 | 58 t / 166 st<br>3/3 blocks | 27 t / 76 st<br>3/3 blocks | – | – | 788 · 696 st / 134 · 130 st |
| q8 KV4-rot | 58 t / 165 st<br>3/3 blocks | 27 t / 74 st<br>3/3 blocks | – | – | – / – |
| q8 RTN-W3 | 58 t / 174 st<br>3/3 blocks | 27 t / 81 st<br>3/3 blocks | – | – | – / – |
| q4 / q8 KV4 (unrotated) | – | – | – | – | 788 · 787 / 788 · 788 st; q8 ALFWorld 134 · 134 st |
| q14 GPTQ-W4 | – | – | – | – | 788 · 471 st / – |

Table 3: Completed crossed-intervention coverage (t = tasks, st = states, b = complete/found blocks; BFCL and ALFWorld: items · states, one run). Bank U: blocks completed and state rows per domain; blocks f01–f03 are analysed (Appendix F.2). "holdout": the cell is in the telecom holdout (Appendix C). Qwen3-32B has end-to-end runs only; Llama-3.1-8B stopped at its fork gate (Limitations); the q14 (Qwen3-14B) BFCL row is a within-family scale check. q4 RTN-W3 has no complete Bank P block and is omitted. The telecom development results (Appendix C.1) use the 6 blocks listed. †The BFCL base category of q4 GPTQ-W3 was dropped by the QQ no-op rule (Appendix F.4). Only completed, admitted panels are counted.

| Config. (models) | Method, weights | Group, grid | Act-order | Calibration | KV cache / activations / rotation |
|---|---|---|---|---|---|
| GPTQ-W4 Q (q4, q8) | GPTQ, W4; damp 0.01, block 128, true-sequential | 128, asym. | yes | draw c1: 57 BF16 agent-trajectory states (39 retail, 18 airline calibration tasks, disjoint from the 85 dev tasks), one state per task, full context (median 6041 / 6310 tok.); 64,000 activation rows (1122–1123 per state); draw seed 20260729 | KV BF16, act. BF16, no rotation |
| GPTQ-W4 Q′ (q4, q8) | as Q | 128, asym. | yes | draw c2, seed 20260730: same 57 tasks and trajectories, re-drawn decision state and rows (5/57 states shared with c1); not disjoint | as Q |
| AWQ-style W4 (q4, q8) | AWQ-style: per-linear scale $\lvert\bar{x}\rvert^{\alpha}$, 20-point $\alpha$ grid, scale folded into weight, then RTN; no clip search | 128, asym. | n/a (GPTQ only) | draw c1 as GPTQ-W4 Q; $\alpha$ fit on a 4096-row sample (code default; per-run value not recorded) | as Q |
| GPTQ-W3$^{s}$ Q / Q′ (q4, q8) | GPTQ, W3 | 128, asym. | yes | c1 / c2 as W4 | as Q |
| RTN-W3$^{s}$ (q4, q8) | RTN, W3 | 128, asym. | n/a | none | as Q |
| KV8 (q4, q8) | BF16 weights | – | – | none | INT8 K and V, asym. min–max per (token, KV head) over head_dim, no clip, no rotation; fake quant in fp32 |
| KV4-rot$^{s}$ (q4, q8) | BF16 weights | – | – | none | INT4 K and V, same grid as KV8; online normalized Hadamard on head_dim: q and k rotated, V rotated and rotated back |
| QuaRot-W4A4 (q8) | random-Hadamard rotation (seed 0) + GPTQ W4, MSE clip search | per out-channel, sym. | no | WikiText-2 train, 128 × 2048 tok., seed 0 | A4: sym. per-token dynamic, clip 0.9, inputs of qkv, o, gate/up, down, online Hadamard (o, down); KV16, per-head q/k Hadamard after RoPE |
| SpinQuant-W4A4 (q8) | learned $R_1$, per-layer $R_2$ (100 steps, lr 1.5, A4 in training) + GPTQ W4 as QuaRot | per out-channel, sym. | no | weights and rotations: WikiText-2, 128 × 2048, seed 0 | as QuaRot-W4A4 |
| KIVI-2 / KIVI-4 (q8) | BF16 weights | – | – | none | K per channel, V per token, group 32, asym.; last 128 tokens full precision |
| Mistral GPTQ-W4 / RTN-W4 (m24) | GPTQ W4 / RTN W4 | 128, asym. | yes / n/a | GPTQ: draw c1, 57 states, 64,000 rows, seed 20260729 (no c2 draw) | as Q |

Table 4: Quantization configuration of every checkpoint configuration, from the checkpoint metadata. AWQ-style W4 is AWQ-style (scale folded into the weight, then RTN, no clip search). Q′ re-draws the decision state and activation rows from the same 57 calibration trajectories as Q. The W4A4 configurations keep a 16-bit KV cache and calibrate on WikiText-2, so a QuaRot-minus-GPTQ contrast also changes the calibration source. One q8 calibration context (c1) is marked truncated. All checkpoints are fake-quantized and stored in BF16; embeddings, `lm_head` and norms stay BF16. $^{s}$stress cell.

## B Complete Upgrade Effects and Their Interpretation

This appendix supports §4: replacing a W4 agent's action produces both rescues and damages, and a gain measured for one upgrade mode cannot be reused as the gain of another.

Table 5 gives every cell of Figure 2a, including the stress cells, with new-task intervals. Table 6 gives $G$, $H$, $F$ and their paired differences for every cell, and Table 7 gives $F - G$, $F - H$ and the $(\Delta_{\mathrm{Q}}, \Delta_{\mathrm{B}})$ summaries.

### B.1 The Two Illustrative Rescue Patterns

Everything in this section is exploratory and post hoc. It uses the four outcomes of each divergent state in the order BB, BQ, QB, QQ; $k$ is the number of successful arms.

**Joint rescue and rescue undone against the reference.** A label-permutation reference for the two named patterns keeps $k$ for every state and permutes the four outcome labels uniformly: `1000` then has probability $1/4$ at $k = 1$, and rescue undone (`0100` or `0110`) has $1/4$ at $k = 1$ plus $1/6$ at $k = 2$, so the reference shares are $P(k{=}1)/4$ and $P(k{=}1)/4 + P(k{=}2)/6$, computed on the same states and bootstrap draws as the observed shares (Table 8). In the $\tau^2$-bench W4 pool, joint rescue exceeds the reference by $+0.47$ pp on the fixed task pool ($[+0.06, +0.87]$), but the new-task interval spans zero ($[-0.33, +1.37]$); rescue undone is less frequent than the reference ($-0.99$ $[-1.58, -0.42]$; new tasks $[-2.19, +0.27]$). The excess of joint rescue is concentrated in the q8 cells, and no group shows a new-task excess of rescue undone. Because the reference keeps $k$ and does not model execution, a share near it says that the arrangement is about as common as a random arrangement of the same number of successes, not that the outcomes are noise or that the continuation is irrelevant. The observed shares in the W4 pool, with both intervals, are 4.0% for joint rescue (fixed pool [3.5, 4.5]; new tasks [2.7, 5.3]) and 4.8% for rescue undone ([4.1, 5.4]; [3.2, 6.5]).

### B.2 Task fragility

All analyses in this subsection are exploratory and post hoc.

**The consequences recur on the same tasks.** We split the 20 nonce blocks into two halves by an outcome-independent hash of the block name, ranked tasks by turnover on one half and measured the other half (Appendix Table 9). The 17 most fragile tasks (top 20%) carry 41.0% (crossed [26.6, 55.9]) and 42.3% ([27.0, 57.5]) of the held-out turnover in the two directions, against 20% if fragility did not persist. Their turnover is 38.7%, while the other 68 tasks still turn over 13.7%. Both groups have rescues and damages; neither

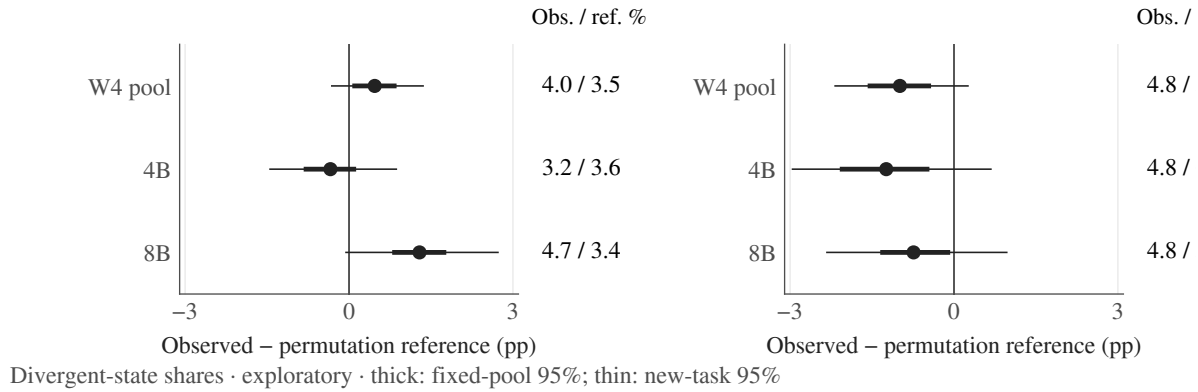



Figure 5: Joint rescue and rescue undone relative to label permutation on W4 divergent states. Thick lines show fixed-pool 95% intervals; thin lines show new-task 95% intervals. The right columns give observed and reference shares. These exploratory comparisons test enrichment, separately from the observed pattern counts in the main text. Values: Table 8.

| Cell | units | div. | $r$ | $d$ | $M$ | $G_{\text{all}}$ [fixed pool] | [new tasks] |
|---|---|---|---|---|---|---|---|
| q4 GPTQ-W4 | 85×20 | 1102 | 6.8 | 8.7 | 15.5 | −1.9 [−3.7, −0.1] | [−6.8, +2.7] |
| q4 AWQ-style W4 | 85×20 | 1439 | 9.8 | 10.1 | 19.9 | −0.3 [−2.2, +1.6] | [−5.4, +5.2] |
| q8 GPTQ-W4 | 85×20 | 1473 | 10.4 | 7.5 | 17.9 | +2.9 [+1.4, +4.4] | [−2.9, +9.2] |
| q8 AWQ-style W4 | 85×20 | 1562 | 11.4 | 9.5 | 20.9 | +1.9 [−0.6, +4.3] | [−3.9, +8.1] |
| W4 pool | 85×20 | | 9.6 | 9.0 | 18.6 | +0.7 [−0.3, +1.6] | [−2.1, +3.4] |
| q4 KV8 | 85×22 | 593 | 3.1 | 3.4 | 6.5 | −0.3 [−1.7, +1.0] | [−3.0, +2.3] |
| q8 KV8 | 85×22 | 855 | 4.3 | 3.7 | 8.1 | +0.6 [−0.4, +1.6] | [−1.9, +3.0] |
| q4 GPTQ-W3$^s$ | 85×3 | 232 | 9.0 | 10.2 | 19.2 | −1.2 [−7.0, +4.7] | [−8.6, +5.9] |
| q4 KV4-rot$^s$ | 85×3 | 233 | 11.8 | 7.1 | 18.8 | +4.7 [−4.1, +13.5] | [−3.1, +12.9] |
| q8 GPTQ-W3$^s$ | 85×3 | 242 | 10.2 | 8.2 | 18.4 | +2.0 [−12.5, +16.4] | [−6.3, +11.8] |
| q8 KV4-rot$^s$ | 85×3 | 239 | 11.8 | 8.6 | 20.4 | +3.1 [−14.5, +20.8] | [−5.9, +12.9] |
| q8 RTN-W3$^s$ | 85×3 | 255 | 3.1 | 3.5 | 6.7 | −0.4 [−3.8, +3.0] | [−4.7, +3.9] |

Table 5: Replacing Q's first tool-structure-divergent action while Q continues, $\tau^2$-bench retail/airline panel (pp). Denominator: the full pool of tasks × complete blocks; units without a Bank P state count as zero. div. = Bank P states. $G_{\text{all}} = r - d$ is a local replacement effect, *not* the end-to-end success difference (Appendix A.2). Fixed pool: block bootstrap (≥10 blocks) or $t$ over block means; new tasks: crossed task×block bootstrap. $^s$stress cell, 3 blocks.

group's net gain is resolved under the crossed interval. This replicates fragility on other nonces of the same tasks; it does not predict which new tasks are fragile.

**Split-half fragility (Table 9).** The halves are assigned by an outcome-independent hash of the block name (sha256 of a fixed prefix and the block name, mod 2), written to a file before this post hoc analysis loaded outcomes; fragility is a task's mean turnover over the four W4 cells and the defining half's blocks, and ties at the top-20% threshold are broken by a hash. In the reverse direction (rank on the second half, estimate on the first), the 17 fragile tasks turn over 39.0% ($t$ [34.2, 43.9]) and carry 42.3% of the held-out turnover, the other tasks 13.3%; the fragile tasks' net gain is +3.7 pp ($t$ [+0.6, +6.9]; crossed [−5.0, +12.4]) and the rest's +0.5. Ranking tasks by BF16 success instead (tertiles on one half) also separates turnover: the hardest tertile turns over 6.7–10.0% and the others 25.4–27.3%, with no net gain resolved under the crossed interval.

| Cell | states | $G$ (action) | $H$ (continuation) | $F$ (joint) | $G - H$ |
|---|---|---|---|---|---|
| | | $-2.9$ [$-5.7$, $-0.2$] | $+0.7$ [$-1.6$, $+3.0$] | $-2.5$ [$-5.3$, $+0.2$] | $-3.6$ [$-6.6$, $-0.6$] |
| q4 GPTQ-W4 | 1102 | [$-10.3$, $+4.2$] | [$-4.0$, $+6.0$] | [$-9.9$, $+4.7$] | [$-11.2$, $+3.6$] |
| | | $-0.3$ [$-2.6$, $+1.9$] | $+0.9$ [$-1.1$, $+2.9$] | $-0.8$ [$-3.3$, $+1.6$] | $-1.3$ [$-3.0$, $+0.6$] |
| q4 AWQ-style W4 | 1439 | [$-6.5$, $+6.1$] | [$-3.7$, $+5.5$] | [$-7.4$, $+5.8$] | [$-7.7$, $+5.1$] |
| | | $+3.3$ [$+1.6$, $+5.0$] | $+0.4$ [$-1.2$, $+1.9$] | $+5.4$ [$+3.5$, $+7.2$] | $+2.9$ [$+1.3$, $+4.7$] |
| q8 GPTQ-W4 | 1473 | [$-3.3$, $+10.5$] | [$-3.8$, $+4.9$] | [$-2.0$, $+13.0$] | [$-3.2$, $+9.3$] |
| | | $+2.1$ [$-0.6$, $+4.7$] | $-1.3$ [$-3.5$, $+0.9$] | $+0.6$ [$-1.5$, $+2.7$] | $+3.4$ [$+1.5$, $+5.2$] |
| q8 AWQ-style W4 | 1562 | [$-4.1$, $+8.7$] | [$-7.1$, $+4.6$] | [$-6.2$, $+7.3$] | [$-1.6$, $+8.6$] |
| | | $+0.5$ [$-0.7$, $+1.7$] | $+0.2$ [$-1.3$, $+1.6$] | $+0.7$ [$-0.6$, $+2.1$] | $+0.4$ [$-0.8$, $+1.7$] |
| W4 pool | 5576 | [$-2.9$, $+3.8$] | [$-2.7$, $+2.8$] | [$-3.1$, $+4.4$] | [$-3.1$, $+3.7$] |
| | | $-1.0$ [$-5.3$, $+3.2$] | $-2.5$ [$-4.5$, $-0.5$] | $-1.2$ [$-4.7$, $+2.3$] | $+1.5$ [$-3.1$, $+6.0$] |
| q4 KV8 | 593 | [$-9.4$, $+7.3$] | [$-6.4$, $+1.1$] | [$-9.3$, $+7.7$] | [$-7.1$, $+9.7$] |
| | | $+1.3$ [$-0.9$, $+3.4$] | $-0.4$ [$-2.1$, $+1.5$] | $+2.5$ [$+0.0$, $+5.1$] | $+1.6$ [$-0.6$, $+3.9$] |
| q8 KV8 | 855 | [$-4.0$, $+6.9$] | [$-3.5$, $+2.9$] | [$-3.5$, $+8.1$] | [$-3.8$, $+7.2$] |
| | | $+0.1$ [$-2.8$, $+3.0$] | $-1.4$ [$-2.7$, $-0.1$] | $+0.6$ [$-1.8$, $+3.1$] | $+1.6$ [$-1.2$, $+4.1$] |
| KV8 pool | 1448 | [$-5.3$, $+5.6$] | [$-3.7$, $+0.9$] | [$-4.5$, $+5.8$] | [$-3.8$, $+6.6$] |
| | | $-1.3$ [$-7.7$, $+5.0$] | $+4.3$ [$-10.7$, $+19.4$] | $+1.3$ [$-1.9$, $+4.4$] | $-5.6$ [$-15.0$, $+3.7$] |
| q4 GPTQ-W3$^s$ | 232 | [$-9.7$, $+6.6$] | [$-6.6$, $+14.5$] | [$-6.8$, $+9.4$] | [$-15.0$, $+4.0$] |
| | | $+5.2$ [$-4.5$, $+14.9$] | $+9.4$ [$+5.1$, $+13.7$] | $+12.4$ [$+11.0$, $+13.9$] | $-4.3$ [$-18.3$, $+9.8$] |
| q4 KV4-rot$^s$ | 233 | [$-3.4$, $+14.2$] | [$+2.1$, $+17.0$] | [$+3.0$, $+21.6$] | [$-13.5$, $+7.0$] |
| | | $+2.1$ [$-13.3$, $+17.2$] | $+0.4$ [$-7.4$, $+8.2$] | $+2.1$ [$-7.4$, $+11.4$] | $+1.7$ [$-9.1$, $+12.3$] |
| q8 GPTQ-W3$^s$ | 242 | [$-6.7$, $+12.1$] | [$-7.2$, $+8.1$] | [$-6.6$, $+11.8$] | [$-7.7$, $+11.2$] |
| | | $+3.3$ [$-15.4$, $+21.9$] | $-0.8$ [$-18.2$, $+16.3$] | $+4.6$ [$-16.0$, $+25.0$] | $+4.2$ [$+2.5$, $+5.8$] |
| q8 KV4-rot$^s$ | 239 | [$-6.6$, $+13.8$] | [$-11.2$, $+8.5$] | [$-6.6$, $+15.9$] | [$-3.3$, $+12.2$] |
| | | $-0.4$ [$-3.8$, $+3.0$] | $+15.3$ [$+0.7$, $+29.9$] | $+18.4$ [$+16.7$, $+20.1$] | $-15.7$ [$-27.5$, $-3.9$] |
| q8 RTN-W3$^s$ | 255 | [$-4.7$, $+3.9$] | [$+5.9$, $+25.9$] | [$+9.4$, $+28.2$] | [$-24.7$, $-7.1$] |

Table 6: Upgrade gains on divergent states (pp). In each two-line cell, the first line gives the estimate and fixed-pool 95% interval; the second gives the new-task crossed 95% interval. $G$, $H$ and $F$ compare action-only, continuation-only and joint replacement with QQ. Fixed-pool intervals use a block bootstrap over 20–22 blocks, or a $t$ interval over three blocks for $^s$stress configurations. Pools average within-configuration estimates equally; stress configurations are separate.

| Cell | $F - G$ | $F - H$ | off-diag. % | rev. % | mean $I$ |
|---|---|---|---|---|---|
| q4 GPTQ-W4 | $+0.5$ $[-1.2, +2.2]$<br>$[-4.7, +6.1]$ | $-3.2$ $[-6.4, +0.2]$<br>$[-11.1, +4.4]$ | 25.0 | 2.6 | $+0.3$ |
| q4 AWQ-style W4 | $-0.4$ $[-2.4, +1.5]$<br>$[-6.2, +5.4]$ | $-1.7$ $[-3.6, +0.4]$<br>$[-8.7, +4.7]$ | 29.3 | 3.1 | $+1.3$ |
| q8 GPTQ-W4 | $+2.1$ $[+0.2, +4.0]$<br>$[-3.0, +7.1]$ | $+5.0$ $[+3.2, +6.8]$<br>$[-2.0, +12.5]$ | 22.9 | 2.7 | $-1.7$ |
| q8 AWQ-style W4 | $-1.5$ $[-4.2, +0.8]$<br>$[-7.6, +4.2]$ | $+1.9$ $[-0.1, +3.8]$<br>$[-3.8, +7.5]$ | 30.5 | 4.2 | $+0.3$ |
| W4 pool | $+0.2$ $[-1.0, +1.2]$<br>$[-2.8, +3.1]$ | $+0.5$ $[-1.0, +2.1]$<br>$[-3.5, +4.5]$ | 26.9 | 3.2 | $+0.0$ |
| q4 KV8 | $-0.2$ $[-3.5, +3.7]$<br>$[-5.7, +5.3]$ | $+1.3$ $[-2.5, +4.9]$<br>$[-6.7, +9.5]$ | 15.7 | 1.2 | $-2.4$ |
| q8 KV8 | $+1.2$ $[-0.5, +3.2]$<br>$[-2.5, +5.0]$ | $+2.8$ $[+0.3, +5.5]$<br>$[-3.1, +8.7]$ | 14.9 | 1.2 | $-1.5$ |
| KV8 pool | $+0.5$ $[-1.6, +2.8]$<br>$[-2.8, +3.9]$ | $+2.1$ $[-0.2, +4.2]$<br>$[-3.0, +7.3]$ | 15.3 | 1.2 | $-1.9$ |
| q4 GPTQ-W3$^s$ | $+2.6$ $[-0.6, +5.8]$<br>$[-5.2, +10.5]$ | $-3.0$ $[-15.1, +9.1]$<br>$[-12.2, +6.6]$ | 31.5 | 3.9 | $+1.7$ |
| q4 KV4-rot$^s$ | $+7.3$ $[-2.0, +16.5]$<br>$[-1.3, +17.1]$ | $+3.0$ $[-1.9, +8.0]$<br>$[-6.2, +12.2]$ | 33.0 | 1.7 | $+2.1$ |
| q8 GPTQ-W3$^s$ | $+0.0$ $[-11.0, +11.1]$<br>$[-9.1, +9.1]$ | $+1.7$ $[-0.1, +3.4]$<br>$[-6.8, +11.0]$ | 23.1 | 4.5 | $+0.4$ |
| q8 KV4-rot$^s$ | $+1.3$ $[-1.8, +4.3]$<br>$[-7.2, +9.9]$ | $+5.4$ $[+0.8, +10.0]$<br>$[-2.5, +14.6]$ | 31.0 | 4.6 | $-2.1$ |
| q8 RTN-W3$^s$ | $+18.8$ $[+13.8, +23.9]$<br>$[+10.2, +27.8]$ | $+3.1$ $[-13.0, +19.2]$<br>$[-7.1, +13.3]$ | 29.0 | 0.4 | $-3.5$ |

Table 7: Paired mode contrasts and outcome patterns on the states of Table 6. Two-line cells give fixed-pool and new-task 95% intervals in that order. Off-diag.: $\Delta_{\mathrm{Q}} \neq \Delta_{\mathrm{B}}$; rev.: $\Delta_{\mathrm{Q}} = -\Delta_{\mathrm{B}} \neq 0$; interaction $I = \Delta_{\mathrm{Q}} - \Delta_{\mathrm{B}}$. Intervals are unadjusted. The four core W4 $G - H$ tests form a Holm family: q4 GPTQ-W4 has adjusted $p = .069$; all four new-task intervals cover zero. A positive $G - H$ compares two upgrades, not either upgrade against QQ.

**A. Joint rescue**

| Group | states | obs. | ref. | obs.−ref. [fixed pool] | [new tasks] |
|---|---|---|---|---|---|
| $\tau^2$-bench W4 pool | 5576 | 3.96 | 3.48 | $+0.47$ [$+0.06$, $+0.87$] | [$-0.33$, $+1.37$] |
| q4 W4 cells | 2541 | 3.22 | 3.56 | $-0.34$ [$-0.83$, $+0.13$] | [$-1.46$, $+0.88$] |
| q8 W4 cells | 3035 | 4.69 | 3.40 | $+1.29$ [$+0.79$, $+1.78$] | [$-0.07$, $+2.74$] |
| $\tau^2$-bench KV8 pool | 1448 | 2.99 | 2.17 | $+0.81$ [$+0.10$, $+1.52$] | [$-0.51$, $+2.19$] |
| Telecom q8 W4 pool | 957 | 3.24 | 3.94 | $-0.70$ [$-1.46$, $+0.05$] | [$-2.47$, $+1.15$] |
| Same-precision W4 (Q′ as B) | 784 | 2.27 | 3.50 | $-1.23$ [$-2.47$, $-0.04$] | [$-2.88$, $+0.51$] |
| BF16, same blocks | 784 | 4.96 | 3.53 | $+1.43$ [$-0.40$, $+3.42$] | [$-0.90$, $+4.26$] |
| Mistral, own BF16 states$^m$ | 464 | 3.4 | 2.9 | $+0.5$ [$-2.4$, $+3.5$] | [$-2.0$, $+3.6$] |

**B. Rescue undone**

| Group | states | obs. | ref. | obs.−ref. [fixed pool] | [new tasks] |
|---|---|---|---|---|---|
| $\tau^2$-bench W4 pool | 5576 | 4.80 | 5.78 | $-0.99$ [$-1.58$, $-0.42$] | [$-2.19$, $+0.27$] |
| q4 W4 cells | 2541 | 4.77 | 6.01 | $-1.24$ [$-2.09$, $-0.45$] | [$-2.97$, $+0.69$] |
| q8 W4 cells | 3035 | 4.82 | 5.56 | $-0.74$ [$-1.35$, $-0.07$] | [$-2.34$, $+0.98$] |
| $\tau^2$-bench KV8 pool | 1448 | 2.31 | 4.58 | $-2.28$ [$-2.98$, $-1.58$] | [$-3.62$, $-0.98$] |
| Telecom q8 W4 pool | 957 | 4.18 | 5.13 | $-0.95$ [$-1.70$, $-0.20$] | [$-2.63$, $+0.61$] |
| Same-precision W4 (Q′ as B) | 784 | 5.77 | 5.75 | $+0.02$ [$-2.33$, $+2.25$] | [$-2.66$, $+2.92$] |
| BF16, same blocks | 784 | 3.82 | 6.14 | $-2.32$ [$-4.46$, $-0.19$] | [$-4.68$, $+0.17$] |
| Mistral, own BF16 states$^m$ | 464 | 4.0 | 4.7 | $-0.7$ [$-2.0$, $+0.7$] | [$-3.0$, $+1.9$] |

Table 8: Joint rescue and rescue undone against the conditional label-permutation reference (keeps the number $k$ of successful branches per state), % of divergent states; exploratory, post hoc. 95% intervals: fixed pool (block bootstrap for the 20- and 22-block $\tau^2$-bench pools, $t$ over per-block ratios for fewer than 10 blocks: 6, or 5 for Mistral), new tasks (crossed task×block). Same-precision rows each fork at their own first divergence. $^m$States whose actions differ, on BF16 trajectories (BB is the BF16 original), five blocks; not comparable with Bank P rows.

| Tasks | | $r$ | $d$ | $M$ | $G_{\text{all}}$ |
|---|---|---|---|---|---|
| fragile | est. | 21.1 | 17.6 | 38.7 | $+3.4$ |
| (top 20%) | $t$ | [18.6, 23.5] | [14.3, 21.0] | [34.9, 42.6] | [$-1.0$, $+7.9$] |
| | crossed | [14.8, 28.1] | [12.4, 23.3] | [32.3, 45.7] | [$-7.1$, $+13.7$] |
| rest | est. | 6.5 | 7.2 | 13.7 | $-0.7$ |
| | $t$ | [5.2, 7.8] | [6.2, 8.3] | [12.5, 14.9] | [$-2.7$, $+1.2$] |
| | crossed | [4.4, 9.0] | [4.8, 9.9] | [10.2, 17.5] | [$-4.1$, $+2.4$] |

Table 9: Repeatable task fragility (exploratory, post hoc; four $\tau^2$-bench W4 cells, equal weight). Tasks are ranked by turnover on one hash-defined half of the 20 nonce blocks and measured on the other half (pp of units; 95% intervals, $t$ over held-out blocks and crossed task×block; direction A to B shown). A replication on other nonces of the same tasks, not a prediction for new tasks. Reverse direction: text.

# C Telecom Holdout and Full Baseline Replay

This appendix supports the holdout statement of §4: the net estimate is negative, the frozen readings classify it as inconclusive because they had no negative category, and both pre-registered secondary checks passed.

The protocol was frozen before any holdout unit was run, and its hash was recorded in the split manifest. It fixes the following design:

- tasks, cells, weights and nonce blocks;
- two fixed halves of six blocks each;
- an outcome-blind repair rule, with reserve blocks used only for infrastructure repair;
- the decisional crossed interval (two-way ANOVA moments, normal 90%);
- the readings: positive if $L > 0$, above 3 pp if $L > 3$, small if the interval lies inside $\pm 3$ pp, otherwise inconclusive;
- the secondary families S1 (both $r > 0$ and $d > 0$) and S2 (persona-stratified permutation, 10,000 draws, no outcome screen);
- six stratum families, with Holm correction within each.

The analysis script refuses holdout data unless the hash matches. Two task-cell pairs were excluded as structural, and the complete-case analysis is identical.

**Full replay of the shared baseline.** QQ originally comes from the recorded episode, whereas the other arms are re-executed. An exploratory excess of only-QQ successes in two q8 cells prompted a direct audit. The initial hash sample matched on 135 of 1202 divergent states. A separately frozen audit then replayed all 1202, checking the original checkpoint, serving configuration and numerical canary before each run. Every replay matched the trajectory, prompts, termination, outcome, budget counters, database and environment state. The original and re-executed arms also use the same official success rule. This verifies the QQ baseline; it does not explain why only QQ succeeds on particular tasks.

**Structural exclusions.** Re-including the two structurally excluded task–cell pairs in all twelve blocks (post hoc sensitivity, frozen estimator) with every unit set to $-1$ or $+1$ gives point estimates of $G_{\text{all}}$ from $-6.68$ $[-10.35, -3.01]$ to $-2.27$ $[-5.96, +1.43]$ pp (crossed 90%). This is a range of point estimates: the interval of the most optimistic scenario includes zero. With their observed units kept in the eleven blocks where they exist and only the two missing units set to $-1$ or $+1$, the estimates are $-4.72$ to $-4.35$ pp, each with a crossed 90% interval below zero. The frozen reading is inconclusive in all 36 combinations. Table 10 gives the confirmatory result. Table 11 gives all six stratum families (Holm within family), which are descriptive. Every level estimate is $\leq 0$, and the Holm-significant levels are negative: step-budget T2 and T3, mobile-data issues ($-9.09$ pp) and the fewest-faults tercile ($-6.37$ pp).

## C.1 Telecom development tasks

On 80 telecom development tasks (6 blocks; fixed-pool 95%), the two q8 holdout cells give $G_{\text{all}} = +2.7$ $[-4.7, +10.1]$ (GPTQ-W4) and $-0.6$ $[-4.9, +3.7]$ pp (AWQ-style W4), against $-5.6$ and $-6.6$ on the holdout; q4 has no telecom development forks.

| Cell (weight) | units | excl. | $G$ | $r$ | $d$ | $\rho_c$ |
|---|---|---|---|---|---|---|
| q4 GPTQ-W4 (1/2) | 396 | 1 | $-3.03$ [$-7.86$, $+1.80$] | $+5.05$ | $+8.08$ | 0.668 |
| q8 GPTQ-W4 (1/4) | 408 | 0 | $-5.64$ [$-9.33$, $-1.94$] | $+4.41$ | $+10.05$ | 0.374 |
| q8 AWQ-style W4 (1/4) | 396 | 1 | $-6.57$ [$-11.98$, $-1.15$] | $+6.31$ | $+12.88$ | 0.644 |
| Weighted ($G_{\text{all}}$) | | | $-4.57$ [$-7.17$, $-1.96$] | $+5.21$ | $+9.77$ | 0.588 |

Table 10: Telecom holdout, pre-registered analysis (34 tasks $\times$ 12 blocks; pp; crossed 90% intervals). Per-cell rows are descriptive; the weighted row is decisional ($G_{\text{all}}$, reading: inconclusive) and S1/S2 (both pass). Excl.: structurally excluded tasks (decision/trajectory count mismatch at search). Alongside interval for the weighted row, $t$ over block means (not decisional): $-4.57$ [$-6.08$, $-3.05$].

| Family | Level | $n$ | $G$ [crossed 90%] | $p$ | Holm $p$ |
|---|---|---|---|---|---|
| step budget[a] | T1 | 138/129/110 | $-0.32$ [$-1.57$, $+0.94$] | .6771 | .6771 |
| | T2 | 124/126/162 | $-2.30$ [$-3.88$, $-0.72$] | .0165 | .0495 |
| | T3 | 123/144/124 | $-1.94$ [$-3.35$, $-0.54$] | .0229 | .0495 |
| persona | Hard | 11 | $-5.99$ [$-12.23$, $+0.26$] | .1147 | .3355 |
| | None | 12 | $-3.65$ [$-7.42$, $+0.13$] | .1118 | .3355 |
| | Easy | 11 | $-4.36$ [$-9.19$, $+0.48$] | .1385 | .3355 |
| BF16 difficulty | 0 | 20 | $-2.35$ [$-4.09$, $-0.62$] | .0257 | .0772 |
| | low | 11 | $-4.54$ [$-8.90$, $-0.19$] | .0860 | .1720 |
| | high | 3 | $-18.75$ [$-38.00$, $+0.50$] | .1091 | .1720 |
| task length | T1 | 9 | $-5.09$ [$-9.81$, $-0.38$] | .0755 | .1511 |
| | T2 | 8 | $-0.78$ [$-4.46$, $+2.90$] | .7270 | .7270 |
| | T3 | 17 | $-6.17$ [$-10.49$, $-1.84$] | .0189 | .0567 |
| issue type | mms | 14 | $-1.15$ [$-3.33$, $+1.04$] | .3880 | .4434 |
| | mobile data | 11 | $-9.09$ [$-14.98$, $-3.20$] | .0111 | .0333 |
| | service | 9 | $-4.40$ [$-10.32$, $+1.52$] | .2217 | .4434 |
| faults | T1 | 19 | $-6.37$ [$-9.91$, $-2.83$] | .0030 | .0091 |
| | T2 | 10 | $-1.94$ [$-7.58$, $+3.69$] | .5704 | .5704 |
| | T3 | 5 | $-2.92$ [$-5.18$, $-0.66$] | .0337 | .0674 |

Table 11: Telecom holdout strata: the six pre-registered families (descriptive; Holm within each family over its three levels). pp. For task-level families, $n$ is the number of tasks and $G$ is $G_{\text{all}}$ over the tasks in the level. [a]Step budget (remaining steps at the divergence) is unit-level: $n$ is the number of divergent units per cell (q4 GPTQ-W4/q8 GPTQ-W4/q8 AWQ-style W4), and $G$ is the level's contribution to $G_{\text{all}}$, so the three levels sum to $G_{\text{all}}$. Tertile cut points were fixed on telecom development tasks.

# D Precision References and Common-State Comparisons

This appendix supports §5: two-sided flips and continuation dependence also appear without more bits; at common states B's extra action value over Q′ is not resolved and equivalence is not shown; and on the eligible states one key-order change that keeps the tool operation identical also changes outcomes (exploratory).

## D.1 Second same-bit-width checkpoint

Per model (final analysis set, $n_{\text{blk}} = 6$), the turnover difference is $\Delta M = +0.6$ [$-3.1$, $+4.3$] pp in q4 and $-2.5$ [$-5.6$, $+0.5$] in q8 (fixed pool, 95%), with crossed intervals [$-5.5$, $+6.9$] and [$-7.3$, $+2.7$]. Within $\pm 3$ pp, equivalence is shown for no per-model $\Delta M$ or $\Delta G_{\text{all}}$; the q8 off-diagonal difference ($+0.40$, 90% [$-2.12$, $+2.92$]) is the only per-model contrast inside the margin. The protocol originally mapped "interval includes 0 and $|\Delta M| \leq 3$ pp" to "similar"; after the interim result had been seen, an amendment replaced that reading by two one-sided tests, which are stricter, and changed no estimate.

**Continuation dependence and the reference intervals.** On their own divergent states (784 per design), the swap's effect depends on the continuation in 26.2% of Q′'s divergent states (fixed pool [20.7, 31.3]), against 25.0% for B ([20.6, 29.0]); each same-precision rate (rescues, damages, continuation dependence) has a fixed-pool 95% lower bound above zero (Amendment A1.3).

Acceptance: every state in both designs passed Q replay and every branch passed the prefix and environment checks; QQ no-op samples passed 265/265 (precision) and 242/242 (same precision), and no directory was quarantined. At the unit level, Cohen's $\kappa$ between the action-side flip indicators of the two substitutions is 0.43 in q4 and 0.61 in q8; at the task level, cross-design agreement at matched noise is 0.765 of within-design replication.

W3 stress check (§5; $n_{\text{blk}} = 3$): every state passed Q replay, QQ no-op samples passed 31/31 and 39/39, and no directory was quarantined. All branches passed the prefix and environment checks except one of 774 in q8 same precision: a Q′Q′ branch that ended in an agent-side context overflow, for which the harness records no prefix comparison; its environment hash and first replayed prompt match the frozen state, and by the protocol it counts as a failure, not an exclusion.

**Equivalence tests (W4).** The difference in turnover is $\Delta M = M - M' = -0.98$ pp (90% CI [$-3.26$, $+1.30$]). No precision-specific excess was detected, but equivalence within $\pm 3$ pp is not shown either (two one-sided tests, $p = 0.067$), and the same holds for the net gain ($\Delta G_{\text{all}} = -1.57$ [$-5.17$, $+2.04$], $p = 0.230$) and, descriptively, for the off-diagonal share ($-1.210$ [$-4.09$, $+1.67$], $p = 0.133$, as the mean of per-block ratios, which the interval and test use; $-1.216$ as the ratio of block sums).

**Units affected by both, or by only one, substitution.** At W4, 107 units flip under both sub-

| Reference (status) | States (denominator) | Action replacing Q's (continuation) | Turn-over | Rescues / damages | Net | Paired comparison [interval; 95% unless marked]; reading |
|---|---|---|---|---|---|---|
| Second W4 checkpoint Q′ | Own first divergences; q4/q8 GPTQ-W4; 1020 task–block units (85 tasks × 6 blocks per model) | B (Q) | 16.8 | 8.0 / 8.7 | −0.7 | $\Delta M = -1.0$; fixed [−3.9, +1.9]; new [−4.5, +2.7]. Unresolved; equivalence not shown. |
| | | Q′ (Q) | 17.7 | 9.3 / 8.4 | +0.9 | |
| Shared ordinary states | 510 BF16-sourced q8 states; 85 tasks × 6 blocks; GPTQ-W4 substitutes | B (Q) | 7.1 | 4.1 / 2.9 | +1.2 | $K(\mathrm{Q}) = +1.0$; two-look 97% $t$ [−4.2, +6.2]; bootstrap [−1.4, +3.5]. Unresolved. |
| | | Q′ (Q) | 8.0 | 4.1 / 3.9 | +0.2 | |
| | | B (B; vs. QB) | 9.0 | 5.3 / 3.7 | +1.6 | $K(\mathrm{B}) = +0.8$; two-look 97% $t$ [−2.2, +3.8]; bootstrap [−2.0, +3.7]. Unresolved; equivalence not shown. |
| | | Q′ (B; vs. QB) | 8.6 | 4.7 / 3.9 | +0.8 | |
| Key order (exploratory; code frozen before outcomes) | 300 q8 GPTQ-W4 first divergent states whose action is one tool call with at least two argument keys, 41 tasks (% of states) | B (Q) | 9.0 | 3.3 / 5.7 | −2.3 | turnover, B minus reversed keys: +2.3 [−2.9, +8.4]; not resolved; no equivalence margin |
| | | Q, keys reversed (Q) | 6.7 | 3.3 / 3.3 | 0.0 | |

Table 12: Reference experiments: values and paired intervals of Figure 3 (per-arm intervals: Appendices D.1 and D.3). Unadjusted common-state intervals and the extension timeline appear in Appendix D.2. Each row group has its own state source and denominator; comparisons are paired only within a group, and the references are never subtracted from B's turnover. Turnover, rescues, damages and net compare the replacing action with Q's action under the same continuation: against QQ when Q continues, and against Q's action followed by B (QB) when B continues. Intervals are those each protocol registers: Q′, fixed pool ($t$ over 6 block means) and new tasks (crossed); common states, $t$ over block means and task-cluster bootstrap, a reading needs both, under the two-look (Pocock) reading that the protocol designates confirmatory and, unadjusted, at the frozen levels (Appendix D.2); key order, the wider of block and crossed bootstrap.

stitutions, 64 only under B and 74 only under the second checkpoint. A within-task block shuffle, which keeps each task's flip propensity, gives 76 [69, 84] units flipping under both, so the two substitutions share affected units beyond what flip-prone tasks explain. The overlap lies in action replacement: over the 1020 W4 units, Cohen's $\kappa$ between the indicators "B's action rescues" and "the second checkpoint's action rescues" is $\kappa_r = 0.60$, and for damages $\kappa_d = 0.54$. Under continuation-only takeover, the affected units are largely substitution-specific: rescue sets under the continuation-only contrast agree with $\kappa = 0.20$, and 70% of continuation-dependent units are continuation-dependent under one substitution only. The difference in action-side net value ($\Delta G_{\mathrm{all}}$) comes only from the substitution-specific units and is not resolved (−1.6 pp; fixed pool [−6.2, +3.0], new tasks [−6.5, +3.6]).

**Full substitution (W4).** Under full substitution B flips more units than Q′ (193 vs. 149; +4.3 [+0.4, +8.3] pp, one of several contrasts); this is a turnover comparison, distinct from the unresolved action-value comparison above.

**W3 stress pair.** The protocol's complete W3 panel (Q GPTQ-W3, Q′ a second W3 calibration draw; q4 and q8, $n_{\mathrm{blk}} = 3$, blocks q3–q5, 510 units) is reported next to W4, never pooled with it. The same-precision substitution produces 17.3 pp of turnover, the BF16 substitution 18.8 pp; off-diagonal shares are 27.6% and 27.3%. $\Delta M = +1.6$ pp (95% [−0.66, +3.80]) is unresolved: the preregistered ±3 pp equivalence test fails ($p = 0.055$); that the 90% interval lies just above zero does not change this. $\Delta G_{\mathrm{all}}$ is equivalent within ±3 pp only in the preregistered pooled population

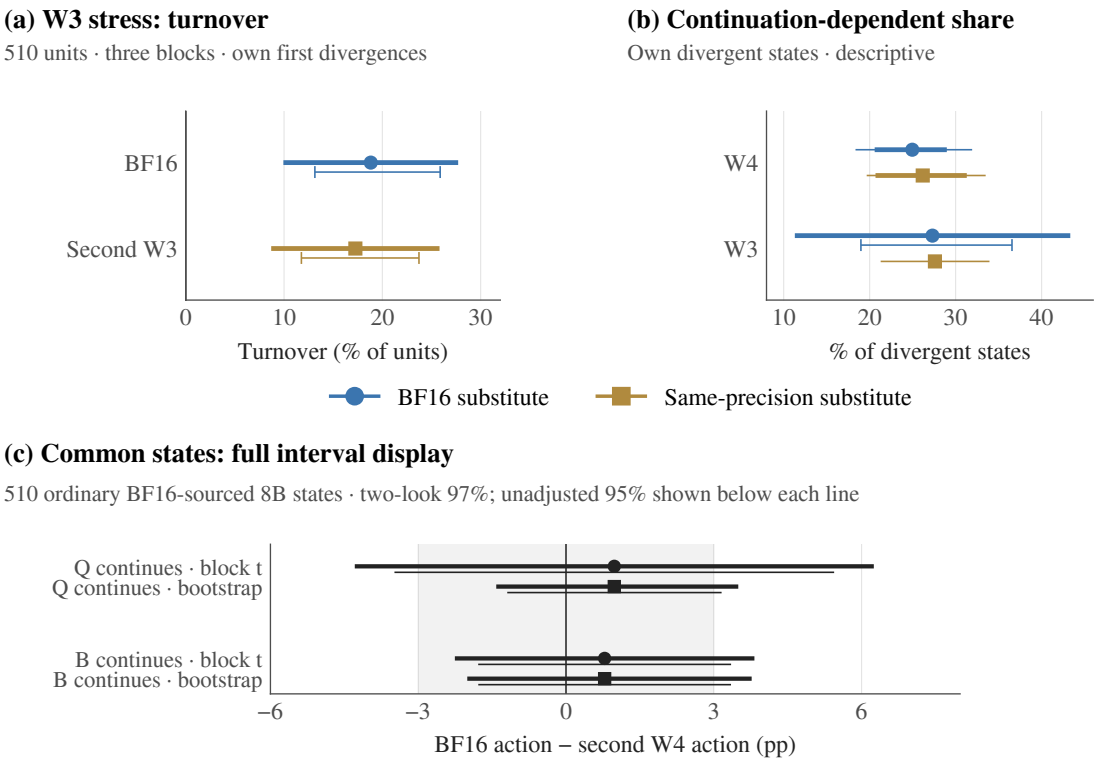



Figure 6: Additional reference results. (a) W3 turnover and (b) continuation-dependent shares: thick fixed-pool and thin new-task 95% intervals. (c) Common-state comparison: block-$t$ and task-bootstrap intervals, with thick two-look 97% and thin unadjusted 95% lines. Shading marks ±3 pp. Each panel has its own state population.

(+1.2, $p = 0.017$); per model it is +3.1 (q4) and −0.8 pp (q8). The off-diagonal difference is −0.288 (ratio of block sums) or −0.307 (mean of per-block ratios, used by the interval and test: 90% [−11.0, +10.4], $p = 0.27$). On the action side ($G$), rescued and harmed task×block units overlap ($\kappa = 0.60, 0.68$); the designs may first diverge at different decisions, so this is unit overlap, not a same-state replication. Of 122 units whose $G$ outcome flips, 62 flip under both substitutions, 34 only under BF16 and 26 only under the other W3 checkpoint: 49% respond to only one. Flip volume, affected units and net upgrade value must be measured separately.

### D.2 Common states: extra value of the BF16 action

**Common-state comparison: details.** Above, each design selects its own divergent states. A comparison pre-registered at three blocks removes that difference: on 510 ordinary BF16-sourced q8 states (85 tasks × 6 blocks, sampled by a fixed rule), the current action comes from B, from Q (GPTQ-W4, first checkpoint) or from Q′ (second checkpoint), and B or Q continues. $K(c) = V(\mathrm{B}, c) - V(\mathrm{Q}', c)$ asks what B's action adds beyond another W4 action. A frozen rule claims value only if both intervals exclude zero (Holm), equivalence only if both lie within ±3 pp (two one-sided tests). The design first ran three blocks, and that first look was unresolved (under Q's continuation $K = -0.4$ pp, 95%: $t$ [−9.3, +8.5], bootstrap [−3.5, +2.7]; under B's +2.4 pp, [−3.5, +8.2], [−0.8, +5.5]). Three more blocks were then added. That decision followed the unresolved first look: a dated amendment fixed the extension and the two-look rule after that look and before any extension run had produced output, and the protocol designates this two-look reading (Pocock levels from a normal approximation: about 97% intervals for a difference and 94% for equivalence) as the confirmatory one. The Pocock constant is not calibrated for small-$k$ $t$ intervals, the task bootstrap or the joint rule, so the two-look reading carries no exact error guarantee. Because the extension was chosen after the first look, even the two-look reading cannot fully protect the error rate conditional on the observed first look (protocol D2.2). We therefore print it next to the unadjusted reading at the frozen levels and claim neither an advantage

nor equivalence unless both agree. After the fixed extension (six blocks), B's action is not shown to be worth more than Q′'s, and equivalence is not established. Under Q's continuation, $K = +1.0$ pp (95%: $t$ $[-3.5, +5.4]$, bootstrap $[-1.2, +3.1]$; two-look 97%: $[-4.2, +6.2]$, $[-1.4, +3.5]$). Under B's, $K = +0.8$ pp (95%: $[-1.8, +3.3]$ for both; two-look 97%: $[-2.2, +3.8]$, $[-2.0, +3.7]$). For equivalence under B's continuation, the ordinary 90% intervals lie within $\pm 3$ pp ($t$ $[-1.2, +2.8]$, bootstrap $[-1.4, +2.9]$) and the 94% intervals of the protocol-designated two-look confirmatory reading do not ($t$ $[-1.6, +3.2]$); because the extension followed the first look, we do not claim equivalence. Without the 26 states at which earlier studies had already produced outcomes, $K$ is $+0.5$ and $+0.8$ pp, also not resolved, and the unadjusted reading no longer shows equivalence under B's continuation either; BF16's own outcome at the extension states was also known in aggregate before the amendment (protocol D2.1). Under Q's continuation, B's action flips $M = 7.1\%$ of outcomes (rescue 4.1, damage 2.9) for a net $\Delta = +1.2$ pp (95%: $t$ $[-1.3, +3.6]$); Q′'s flips 8.0% (4.1, 3.9) for $+0.2$ pp ($[-2.3, +2.7]$; within $\pm 3$ pp by two one-sided tests). A second W4 checkpoint's action departs from Q's at least as often as B's does (agreement with Q's action 53.9% against 55.7%; B with Q′ 57.3%). The interaction of $K$ with the continuation is $-0.2$ pp (95%: $t$ $[-4.5, +4.1]$, bootstrap $[-2.7, +2.4]$; secondary, not resolved). *q4, a separate prospective analysis.* The same design on Qwen3-4B planned three blocks. The frozen quarantine rule removed f01 (below), leaving 170 states. $K = -1.2$ pp under Q's continuation (bootstrap 95% $[-5.9, +3.5]$) and $-2.4$ pp under B's ($[-6.5, +1.2]$), both not resolved. With two blocks the $t$ interval has one degree of freedom. Under Q's continuation the two block means happen to coincide, so it collapses to a point. That is a coincidence of two values, not a precise measurement or zero variance, and the reading rests on the bootstrap. *Identity checks.* Across both models, 1,238 of 1,240 reference and implicit identity checks passed. Both failures are one q4 state. There the substituted action was semantically identical to the original after parsing, but its token sequence differed in the order of keys inside a tool-call argument. The canonical comparison treats such actions as equal, but the chat template renders the order, so the next prompt differed and the continuation could not be a replay. Every check with a token-identical action passed (869/869). As the frozen rule requires, the block was removed.

**Common states, BF16 against Q′, by component.** At the same states and continuation, BF16's extra rescues over Q′ on q8 are $+0.3$ pp and its extra damages $-0.6$ pp, with the net $K = +0.9$ unresolved; each component lies within $\pm 3$ pp under a 90% two one-sided test (exploratory, post hoc; the margin is wide relative to rescue rates of 3–5%). q4 is unresolved.

### D.3 Key-order control

**Design.** Protocol `KEYORDER_PERTURB_PROTOCOL.md` (ko-v1, exploratory control). Eligible states are first tool-structure divergences of q8 GPTQ-W4 (Bank P) where Q's action is a tool call with at least two top-level argument keys, reversing them leaves the parsed call and the environment effects unchanged, and the next prompt tokens differ (300 of 1473 divergent states, 41 tasks). The arm KQ forces the reversed action and lets Q continue; QQ and BQ are reused from Bank P. Eligibility was fixed without outcomes; the analysis code (sha256-recorded) was frozen before any KQ outcome was read. All four tasks passed acceptance (KQ checks 300/300, shadow 300/300, QQ no-op 35/35, no errors or context overflows). Two tasks were restarted with resume during the run (Amendment 2026-10-05g).

**Results.** Conditional on eligible states (pooled ratio; intervals are the wider of block and crossed bootstrap), KQ changes 6.7% [0.8, 14.4] of outcomes and B 9.0% [2.9, 18.2]; KQ has 10 rescues and 10 damages, B 10 and 17, and the flips coincide on 5 rescues and 7 damages. Rescue and damage shares (Figure 3b): KQ 3.3% [0.0, 9.3] and 3.3% [0.4, 8.0]; B 3.3% [0.0, 9.2] and 5.7% [1.5, 11.5]. Over the full 1700-unit pool, KQ contributes 1.2 pp [0.1, 2.8] of turnover against 17.9 pp for the BF16 swap on all divergent states.

## E Selecting Decisions for Signed Benefit

This appendix supports §6. It specifies the frozen trigger, matched random and hindsight references, and the development diagnostics used to choose the rule. The net-benefit tests remain unresolved after correction.

### E.1 The frozen trigger and its headroom

On the first development block, we scored each ordinarily sampled state at the token level (1140 cell $\times$ state rows, 58 flips), teacher-forcing both models on the generated ids of $a_{\mathrm{B}}$ and $a_{\mathrm{Q}}$. We record truncated top-20 entropy, the top-1 margin and a top-20-union KL, not full entropy or KL. All analyses are exploratory, and f01 is development data.

**Uncertainty is associated with divergence.** Q's entropy at the start of its action separates states where B and Q act differently from states where they act identically, with AUC 0.66–0.91 per sub-bank and model. Over all ordinary states, flips also sit at lower Q margins (U1: AUC 0.31 [0.22, 0.40] for q4, 0.24 [0.13, 0.36] for q8; stratified). Among already divergent U1 states, no Q-visible statistic has an interval excluding AUC 0.5; small-subset associations vary across state sources. This development diagnosis motivated the frozen margin rule, whose validation reuses the tasks with fresh nonce blocks. On the secondary validation population, ordinary states from BF16 trajectories (U2B; f02/f03, 1020 rows, 75 outcome-changing states), the frozen rule captures 4, 11 and 35 outcome-changing states against 3.81, 8.35 and 16.07 expected at random ($p = .5404$ and .1995 at 5 and 10%; the 20% excess is resolved), with net gains of $-2$, $-1$ and $+5$ against random $+0.07$, $+1.10$ and $+2.48$.

### E.2 Matched references for the frozen trigger

Table 13 compares the frozen trigger of §6 (frozen rule, unchanged) with two references on the same 1020 f02/f03 U1 rows, which hold 38 rescues and 31 damages. The *matched action-replacement oracle* selects, per cell, exactly as many rows as the frozen rule did, taking only rows with $G > 0$ and filling with $G = 0$ rows when a cell has too few rescues; it never selects a damage. It is hindsight selection, not a policy. At the 10% and 20% budgets every rescue fits.

**A. Frozen trigger and matched hindsight oracle**

| Budget | sel. | oracle r/d/net (pp) | frozen r/d/net (pp) | net ratio |
|---|---|---|---|---|
| 5% | 50 | 34/0/+34 (+3.33) | 6/2/+4 (+0.39) | 4/34 |
| 10% | 114 | 38/0/+38 (+3.73) | 7/5/+2 (+0.20) | 2/38 |
| 20% | 222 | 38/0/+38 (+3.73) | 16/11/+5 (+0.49) | 5/38 |

**B. Frozen trigger against matched-count random selection (Figure 4)**

| Budget | sel. | net vs. random ($p$; Holm) | r/d | turnover vs. random ($p$) |
|---|---|---|---|---|
| 5% | 50 | +4 vs. +0.21 (.0345; .1035) | 6/2 | 8 vs. 3.48 (.0182) |
| 10% | 114 | +2 vs. +0.98 (.4169; .5234) | 7/5 | 12 vs. 7.52 (.0613) |
| 20% | 222 | +5 vs. +2.38 (.2617; .5234) | 16/11 | 27 vs. 14.30 (.0002) |

Table 13: Frozen Q-only trigger against a matched action-replacement oracle (hindsight; same per-cell selection counts), on f02/f03 U1 (1020 rows). Net ratio: frozen net over oracle net (not a share of rescues; the frozen rule selects 6, 7 and 16 of the oracle's rescues). Random: matched-count per-cell random-trigger expectation; $p$: exact one-sided hypergeometric; Holm over the three budgets is applied to the net $p$-values only.

# F Scope Across States, Environments and Models

This appendix answers the scope questions behind Table 1: does the turnover–value distinction extend to ordinary states, environments without an LLM user, another model family and other quantization settings? The simulator controls also delimit the fixed-execution interpretation in §7.

## F.1 Ordinary states and the user simulator

**Ordinary states include agreements.** At ordinary states, replacing Q's action changes the outcome less often than at first tool-structure divergences (a descriptive comparison of different state populations, with no interval on the difference). Over blocks f01–f03 (task-equal; $t$ intervals over block means), $M$ per sub-bank is 7.8–9.0% for the q4 W4 cells in U1 and U2B, 4.7–9.0% for the q8 W4 cells and at most 4.7% for KV8. The net gain lies within $\pm 2.8$ pp (U1, U2B) and $\pm 5.0$ pp (U2Q), and every interval covers zero (Appendix F.2). Identical actions never change the outcome, as determinism implies. Reused arms rest on canonical equality. All 4,114 such reuses were rendered with each model's chat template; 4,113 are token-identical. The remaining pair differs only in JSON key order, and its two cross arms, measured separately, gave the reused outcomes.

**Sensitivity to the user simulator.** Text actions act through the user simulator, and changing the simulator moves $\tau$-bench outcomes by 9–15 points (Seshadri et al., 2026; Jain and Sandhu, 2026). We froze a control before running it: the 39 deduplicated text-difference flips with the same tool structure of block f01 and 39 matched non-flip states. Their BQ and QQ branches were each re-run three times with a different user simulator, Sonnet 5.5, answering through a file relay from Claude Code subagents (reproducibility limits in the Limitations). Judged by the sign of the replicate mean, only 7 of the 39 flips keep their direction (95% CI 0.075–0.335). Six reverse, and 26 vanish (15 both succeed, 2 both fail, 9 ties). Among the matched non-flip states, 11 of 39 become flips (0.15–0.45). Replicates of the same arm disagree in 41% of flip arms and in 21–26% of control arms. **The Bank U text-difference outcome labels did not reproduce reliably under this alternative relay setting.** The control changes three things at once (simulator model, sampling, relay interface), so it cannot say which of them changed the labels.

**Sensitivity of the four-branch results.** Before running, we sampled 80 Bank P states without outcome information (20 per W4 cell, stratified by domain and block) and re-ran all four arms five times each with Sonnet 5.5 as the user, one isolated single-request call per user turn (7896 calls; all integrity checks passed). Per-state arm outcomes track the original single runs (correlation 0.63–0.68). Pooled over cells, $G = -3.0$ $[-10.2, +4.8]$, $H = +5.5$ $[+0.0, +11.2]$ and $G - H = -8.5$ $[-14.8, -2.8]$ pp under Sonnet, against $G - H = -2.5$ $[-11.2, +6.2]$ on the same states under the original simulator (change $-6.0$ $[-16.2, +4.2]$). With 20 states per cell, this control cannot resolve the few-pp contrasts of Table 6 (on this sample even the original simulator gives $G - H = 0.0$ in both q8 W4 cells), so it neither confirms nor reverses the within-cell orderings.

## F.2 Ordinary-state estimates by history source

Table 14 gives the per-state turnover behind Appendix F.1, over blocks f01–f03. Exact action agreement is 0.43–0.83 per cell and sub-bank.

## F.3 Models compared on their own ordinary states

**Mistral population, scoring and coverage.** The Mistral panel applies the same row rule to Mistral's own BF16 trajectories on the 85 retail and airline tasks. Seven BF16 source rows are missing because the source episode ended in an invalid tool call produced by the model itself (airline task 44 in all six blocks, airline task 13 in f01). A dated amendment, fixed from the BF16 source runs before any Mistral fork output existed, excludes both tasks in every block and cell, leaving 83 tasks. The selection was outcome-blind, but BF16 Mistral fails these tasks structurally, so the estimates describe this conditional population, not the 85-task panel, and are not compared with q8 as if the populations were equal. A branch that ended because the harness could not accept the agent's own message (an empty or invalid tool name, or an empty message) is scored as a failure, as in end-to-end evaluation. This rule was added after the Mistral phase-1 results had been read; without it the Mistral primary comparison is not estimable from phase 1 (one complete block). The analy-

| Cell | U1 ($n$=255) | U2B ($n$=255) | U2Q ($n$=60) |
|---|---|---|---|
| q4 GPTQ-W4 | .090 (11/12) | .082 (10/11) | .083 (4/1)* |
| q4 AWQ-style W4 | .082 (7/14) | .078 (7/13) | .083 (3/2) |
| q4 KV8 | .047 (4/8) | .039 (5/5) | .017 (0/1) |
| q8 GPTQ-W4 | .047 (8/4) | .067 (9/8) | .050 (3/0)* |
| q8 AWQ-style W4 | .086 (14/8) | .090 (15/8) | .067 (3/1) |
| q8 KV8 | .035 (5/4) | .031 (6/2) | 0 (0/0) |

Table 14: Bank U, blocks f01–f03: per-state turnover $M$ (rescues/damages), task-equal (one state per task and block; 85/85/20 states per block). $G$ lies within $\pm 0.028$ in U1 and U2B and within $\pm 0.050$ in U2Q, and every interval includes 0. Intervals are $t$ intervals over the three block means and are wide (e.g. q4 GPTQ-W4 U1: .090 [.029, .151]). *GPTQ-W4's U2Q states are its own U1 states, with concordant outcomes on 20/20 duplicates per block; they are counted once in cross-sub-bank totals.

| Cell | blocks | states | $G$ | $H$ | $F$ |
|---|---|---|---|---|---|
| GPTQ-W4 | 6 | 498 | $-0.8$ [$-5.2$, $+3.6$] [$-5.8$, $+4.2$] | $+1.4$ [$-3.0$, $+5.8$] [$-4.0$, $+6.8$] | $+0.0$ [$-4.2$, $+4.2$] [$-6.0$, $+6.0$] |
| RTN-W4 | 5 | 415 | $+0.7$ [$-2.7$, $+4.2$] [$-3.6$, $+5.5$] | $+1.9$ [$-3.8$, $+7.7$] [$-4.6$, $+8.0$] | $+2.4$ [$-2.2$, $+7.0$] [$-3.9$, $+8.4$] |

Table 15: Mistral-Small-3.2-24B on its fixed 83-task conditional population. pp; fixed-pool 95% $t$ intervals, then crossed task×block 95% intervals. Missing argument-JSON errors leave RTN f04 excluded, whereas model-produced invalid tool names are scored as failures under the dated amendment above. None of the three mode intervals excludes zero.

sis without the rule (which leaves such an arm missing and fails its block) and one applying the rule only to data not yet read when it was fixed are reported alongside; both admit the same three blocks for GPTQ − RTN ($+1.2$ [$-13.8$, $+16.2$]). Coverage: GPTQ-W4 has 498 states on six blocks and RTN-W4 415 states on five. Full estimates on the 83 tasks (GPTQ-W4 on six blocks, RTN-W4 on five; Table 15): in GPTQ-W4, B's action rescues $r = 4.6$ [2.4, 6.8] and damages $d = 5.4$ [2.4, 8.5] pp of states ($M = 10.0$ [7.0, 13.1]; fixed-pool 95%), and no mode's net value is resolved ($G = -0.8$ [$-5.2$, $+3.6$], $H = +1.4$ [$-3.0$, $+5.8$], $F = +0.0$ [$-4.2$, $+4.2$]). RTN-W4 is similar ($M = 10.4$ [6.3, 14.4]). The pre-registered contrast GPTQ − RTN is not resolved: $\Delta D_{cc} = +2.2$ [$-3.5$, $+7.8$] pp, $p = .35$, on five blocks, two of which are admitted only by a scoring rule adopted after an interim look (above).

### F.4 Replacement and four-branch estimands on BFCL and ALFWorld (supplementary)

Table 16 gives the replacement quantities of §4 ($r$, $d$, $M$ and $G_{\text{all}}$ over all items, non-divergent items counting as zero) and applies the four-branch analysis to the external benchmarks, from the raw four-arm outcomes on divergent items (one run per item; item bootstrap within benchmark category). The four-branch columns are post hoc and not part of any pre-registered analysis. In the KV4 cells $Y_{\text{QQ}} = 0$ on every state, so $\Delta_{\text{Q}}$ is almost constant and their off-diagonal mass is $H$. In BFCL q4 GPTQ-W3, the base category was dropped whole by the QQ no-op rule, leaving 591 items and 540 states. Off-diagonal shares are 4.7–17.1% in BFCL W4 cells (pool 10.3%) and 32.8–40.0% in q8 ALFWorld W4 (pool 36.4%). The ALFWorld GPTQ-W3 stress cell has $H = +38.5$ and $G - H = -34.6$ [$-43.5$, $-25.4$] pp; KV4 cells reduce to $H$.

### F.5 Quantization methods on shared ordinary states

These scope checks ask how the three upgrade values change across quantization settings. Methods share ordinary BF16-sourced states (U2B); the separate Mistral population is specified in Appendix F.3.

**Interpretation.** The rows below test the range of action-only, continuation-only and joint upgrade values. They report every admitted configuration, including the unquantized path controls. They are descriptive scope checks, not a ranking of quantization algorithms. The configurations differ in calibration pipelines as well as numerical trans-

**A. Full item pool**

| Cell | items | $r$ | $d$ | $M$ | $G_{\text{all}}$ |
|---|---|---|---|---|---|
| BFCL q4 GPTQ-W4 | 788 | 6.3 | 4.3 | 10.7 | $+2.0$ $[-0.3, +4.3]$ |
| BFCL q4 AWQ-style W4 | 788 | 5.7 | 9.5 | 15.2 | $-3.8$ $[-6.5, -1.3]$ |
| BFCL q8 GPTQ-W4 | 788 | 2.9 | 2.3 | 5.2 | $+0.6$ $[-0.9, +2.3]$ |
| BFCL q8 AWQ-style W4 | 788 | 4.1 | 2.7 | 6.7 | $+1.4$ $[-0.3, +3.3]$ |
| BFCL W4 pool | | | | | $+0.1$ $[-1.0, +1.2]$ |
| ALFWorld q8 GPTQ-W4 | 134 | 9.0 | 12.7 | 21.6 | $-3.7$ $[-11.2, +4.5]$ |
| ALFWorld q8 AWQ-style W4 | 134 | 10.4 | 14.2 | 24.6 | $-3.7$ $[-11.9, +4.5]$ |
| ALFWorld W4 pool | | | | | $-3.7$ $[-9.7, +1.9]$ |
| BFCL q4 GPTQ-W3$^s$ | 591 | 6.1 | 3.6 | 9.6 | $+2.5$ $[+0.2, +5.1]$ |
| BFCL q8 GPTQ-W3$^s$ | 788 | 5.2 | 4.1 | 9.3 | $+1.1$ $[-1.0, +3.3]$ |
| BFCL q4 KV4$^s$ | 788 | 0.3 | 0.0 | 0.3 | $+0.3$ $[+0.0, +0.6]$ |
| BFCL q8 KV4$^s$ | 788 | 0.3 | 0.0 | 0.3 | $+0.3$ $[+0.0, +0.6]$ |
| ALFWorld q8 GPTQ-W3$^s$ | 134 | 6.7 | 3.0 | 9.7 | $+3.7$ $[-1.5, +9.7]$ |
| ALFWorld q8 KV4$^s$ | 134 | 0.0 | 0.0 | 0.0 | $+0.0$ $[+0.0, +0.0]$ |

**B. First divergent states**

| Cell | states | $G$ | $H$ | $F$ | $G - H$ | off-diag. % |
|---|---|---|---|---|---|---|
| BFCL q4 GPTQ-W4 | 573 | $+2.8$ $[-0.4, +5.9]$ | $+1.4$ $[-0.7, +3.5]$ | $+4.2$ $[+0.9, +7.2]$ | $+1.4$ $[-1.8, +4.5]$ | 11.7 |
| BFCL q4 AWQ-style W4 | 642 | $-4.7$ $[-7.8, -1.5]$ | $-1.1$ $[-3.6, +1.4]$ | $-1.4$ $[-4.8, +1.7]$ | $-3.6$ $[-6.7, -0.2]$ | 17.1 |
| BFCL q8 GPTQ-W4 | 533 | $+0.9$ $[-1.3, +3.3]$ | $+1.5$ $[+0.0, +3.2]$ | $+2.1$ $[-0.4, +4.6]$ | $-0.6$ $[-3.1, +2.0]$ | 4.7 |
| BFCL q8 AWQ-style W4 | 548 | $+2.0$ $[-0.4, +4.7]$ | $+1.8$ $[+0.0, +3.7]$ | $+3.6$ $[+0.9, +6.5]$ | $+0.2$ $[-2.6, +3.1]$ | 7.8 |
| BFCL W4 pool | | $+0.3$ $[-1.2, +1.8]$ | $+0.9$ $[-0.2, +2.0]$ | $+2.1$ $[+0.3, +3.9]$ | $-0.6$ $[-2.3, +1.0]$ | 10.3 |
| ALFWorld q8 GPTQ-W4 | 115 | $-4.3$ $[-13.3, +5.0]$ | $+7.0$ $[-2.6, +16.7]$ | $+9.6$ $[-0.9, +19.7]$ | $-11.3$ $[-20.9, -1.8]$ | 40.0 |
| ALFWorld q8 AWQ-style W4 | 119 | $-4.2$ $[-13.6, +4.9]$ | $+5.0$ $[-2.6, +12.2]$ | $+4.2$ $[-5.9, +13.7]$ | $-9.2$ $[-18.8, +0.0]$ | 32.8 |
| ALFWorld W4 pool | | $-4.3$ $[-11.1, +2.2]$ | $+6.0$ $[-0.5, +12.6]$ | $+6.9$ $[-2.1, +15.5]$ | $-10.3$ $[-17.7, -2.9]$ | 36.4 |
| BFCL q4 GPTQ-W3$^s$ | 540 | $+2.8$ $[+0.2, +5.5]$ | $+5.2$ $[+2.9, +7.7]$ | $+8.9$ $[+5.7, +12.2]$ | $-2.4$ $[-5.7, +0.7]$ | 13.0 |
| BFCL q8 GPTQ-W3$^s$ | 696 | $+1.3$ $[-1.1, +3.7]$ | $+1.4$ $[-0.1, +2.9]$ | $+3.4$ $[+0.9, +6.0]$ | $-0.1$ $[-2.6, +2.4]$ | 8.3 |
| BFCL q4 KV4$^s$ | 787 | $+0.3$ $[+0.0, +0.6]$ | $+2.9$ $[+1.9, +4.1]$ | $+20.3$ $[+17.6, +23.0]$ | $-2.7$ $[-3.9, -1.5]$ | 20.5 |
| BFCL q8 KV4$^s$ | 788 | $+0.3$ $[+0.0, +0.6]$ | $+1.8$ $[+1.0, +2.7]$ | $+14.5$ $[+12.1, +17.0]$ | $-1.5$ $[-2.5, -0.6]$ | 14.7 |
| ALFWorld q8 GPTQ-W3$^s$ | 130 | $+3.8$ $[-1.5, +9.8]$ | $+38.5$ $[+29.8, +46.9]$ | $+42.3$ $[+33.6, +50.8]$ | $-34.6$ $[-43.5, -25.4]$ | 30.8 |
| ALFWorld q8 KV4$^s$ | 134 | $+0.0$ $[+0.0, +0.0]$ | $+38.1$ $[+30.6, +45.5]$ | $+53.0$ $[+44.8, +61.2]$ | $-38.1$ $[-45.5, -30.6]$ | 26.9 |

Table 16: Replacement and four-branch estimands on BFCL multi-turn and ALFWorld (no LLM user simulator; one run per item; 95% item-bootstrap intervals). All items: rescue $r$, damage $d$, turnover $M$ (% of all items) and net $G_{\text{all}}$ (pp), non-divergent items counting as zero. Divergent items (supplementary, exploratory, post hoc; pp, divergent items): Off-diag.: share with $\Delta_{\text{Q}} \neq \Delta_{\text{B}}$. Pools: equal weight per cell. $^s$Stress and unrotated KV4 cells; in KV4 $Y_{\text{QQ}} = 0$ on every state, so the table reduces to $H$.

formations. KIVI is a replay-compatible variant: each decoding step reconstructs a quantized view from the full-precision cache rather than retaining a compressed cache across turns. Admission and one-time infrastructure re-run rules are recorded in the supplementary panel protocol.

**All admitted ordinary-state upgrade effects.** Table 17 reports all three upgrade modes for every admitted configuration, with both fixed-pool and new-task intervals.

| Cell | blocks | states | $G$ | $H$ | $F$ |
|---|---|---|---|---|---|
| AWQ-style W4 | 3 | 255 | $+2.7$ $[-4.6, +10.1]$<br>$[-2.4, +9.0]$ | $+4.3$ $[-0.1, +8.8]$<br>$[-1.2, +11.0]$ | $+5.5$ $[+1.0, +10.0]$<br>$[-0.8, +12.5]$ |
| GPTQ-W4 | 3 | 255 | $+0.4$ $[-7.0, +7.7]$<br>$[-4.3, +5.9]$ | $+0.8$ $[-5.3, +6.9]$<br>$[-5.9, +7.5]$ | $+3.5$ $[-2.3, +9.4]$<br>$[-3.9, +10.6]$ |
| KIVI-16 path | 3 | 255 | $+0.0$ $[-2.9, +2.9]$<br>$[-2.7, +2.7]$ | $+1.2$ $[-6.6, +8.9]$<br>$[-3.5, +5.9]$ | $+2.4$ $[-0.6, +5.3]$<br>$[-2.4, +7.1]$ |
| KIVI-2 | 2 | 170 | $-2.9$ $[-25.4, +19.5]$<br>$[-8.2, +2.4]$ | $+2.9$ $[-4.5, +10.4]$<br>$[-3.5, +10.0]$ | $+2.9$ $[-34.4, +40.3]$<br>$[-5.9, +10.6]$ |
| KIVI-4 | 3 | 255 | $+0.0$ $[-5.1, +5.1]$<br>$[-3.5, +4.3]$ | $+2.0$ $[-1.4, +5.3]$<br>$[-3.5, +7.5]$ | $+0.8$ $[-8.1, +9.7]$<br>$[-5.9, +7.1]$ |
| KV8 | 3 | 255 | $+1.6$ $[-0.1, +3.3]$<br>$[-1.2, +4.7]$ | $+5.1$ $[+0.6, +9.6]$<br>$[+0.4, +9.8]$ | $+5.5$ $[+1.0, +10.0]$<br>$[+0.0, +10.6]$ |
| QuaRot-W16 path | 3 | 255 | $+0.0$ $[-5.1, +5.1]$<br>$[-3.5, +3.1]$ | $+1.6$ $[-14.5, +17.7]$<br>$[-5.9, +9.4]$ | $+2.4$ $[-8.2, +12.9]$<br>$[-4.3, +9.0]$ |
| QuaRot-W4 | 3 | 255 | $+0.8$ $[-5.3, +6.9]$<br>$[-3.5, +5.9]$ | $+1.6$ $[-1.8, +4.9]$<br>$[-4.3, +7.5]$ | $+4.7$ $[+1.8, +7.6]$<br>$[-2.0, +11.4]$ |
| QuaRot-W4A4 | 3 | 255 | $+1.2$ $[-6.6, +8.9]$<br>$[-4.7, +6.3]$ | $+6.3$ $[-8.4, +21.0]$<br>$[-2.4, +14.1]$ | $+9.4$ $[+1.7, +17.1]$<br>$[+2.0, +16.5]$ |
| SpinQuant-W16 path | 3 | 255 | $-0.4$ $[-2.1, +1.3]$<br>$[-3.1, +2.4]$ | $+2.0$ $[-1.4, +5.3]$<br>$[-2.4, +6.3]$ | $+2.4$ $[-0.6, +5.3]$<br>$[-2.4, +7.1]$ |
| SpinQuant-W4A4 | 3 | 255 | $+3.1$ $[-2.9, +9.2]$<br>$[-2.0, +8.2]$ | $+11.8$ $[+6.7, +16.8]$<br>$[+5.1, +18.4]$ | $+13.7$ $[+9.3, +18.2]$<br>$[+5.5, +21.6]$ |

Table 17: All q8 ordinary BF16-sourced states admitted by the panel rules, including unquantized path controls. Effects in pp; each cell: estimate with fixed-pool 95% $t$ interval, then crossed task$\times$block 95% interval. All three modes use the same admitted states within a row. These descriptive mode intervals are unadjusted. KIVI-2 admits only two blocks after one-time infrastructure re-runs; it is not estimable under the original strict rule. Other cells admit three blocks. No action-only interval excludes zero; several takeover or joint-upgrade intervals do.